MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, BINARY, DIGITAL, KEY CODE NZC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic binary counter module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.267 failures per million operating hours for class I or 1.658 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be NZC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/319-1 for class I and M28787/319-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

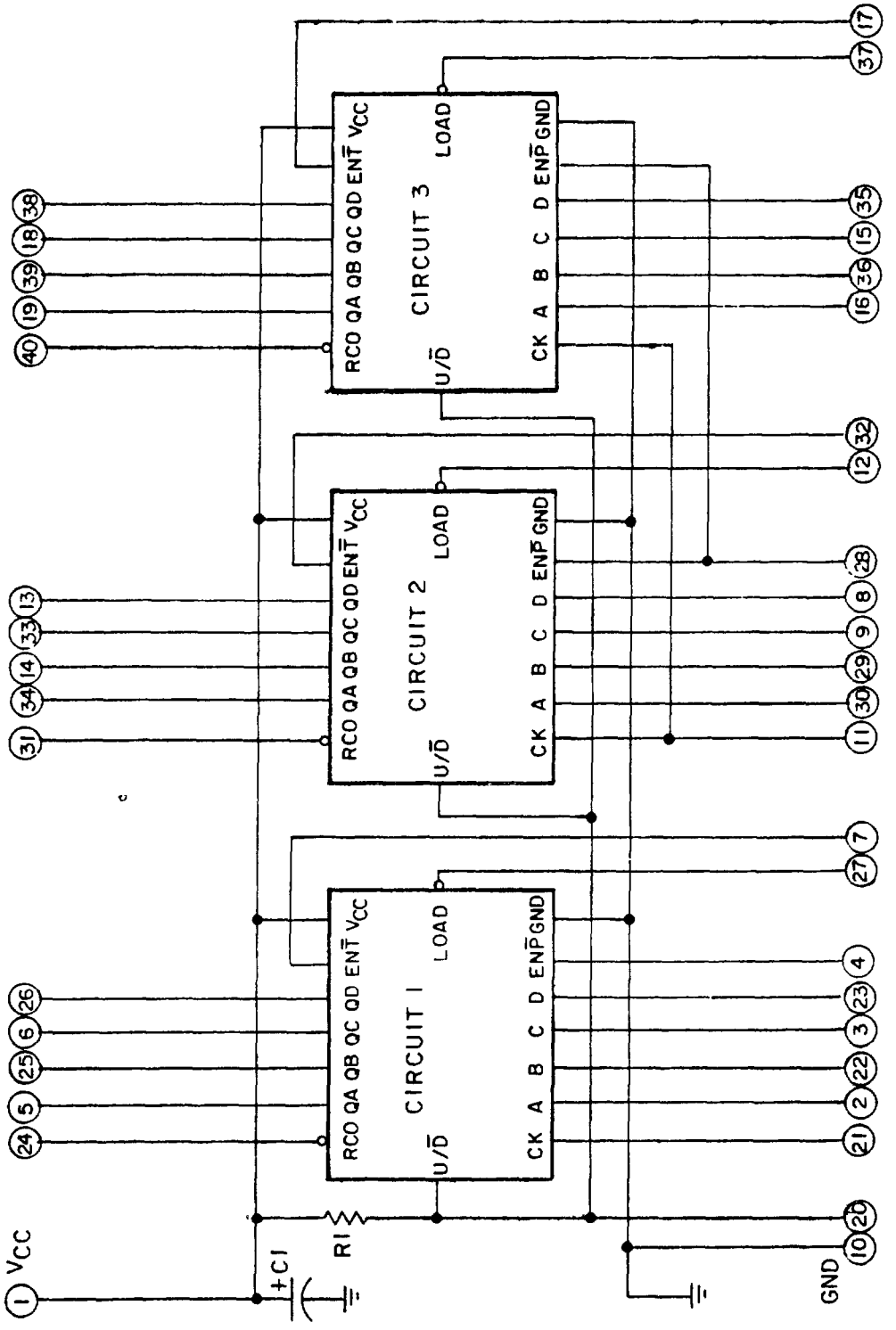
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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NOTES
1. C1 = 2.2 µF ±10%, 50 V.
2. R1 = 10 kΩ ±5%, 1/4 W.

FIGURE 1. NZC module function diagram, binary counter.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B15	Output short circuit current	IV	X	-15	-130	-15	-130	-15	-130	-15	-130	-15	-130	mA
C1-C15	V_{OUT}, low loaded	V	S	10	800	10	800	10	800	10	800	10	800	mV
D1-D5,D10-D13,D18-D21	Input breakdown current	VI	S	0	100	0	100	0	100	0	100	0	100	µA
D6,D9,D14,D15,D22	Input breakdown current	VI	S	0	200	0	200	0	200	0	200	0	200	µA
D7,D16,D23	Input breakdown current	VI	S	0	300	0	300	0	300	0	300	0	300	µA
D8	Input breakdown current	VI	S	-20	100	-20	100	-20	100	-20	100	-20	100	µA
D17	Input breakdown current	VI	S	0	400	0	400	0	400	0	400	0	400	µA
E1-E23	Input clamp diode voltage	VII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
F1-F5,F10-F13,F18-F21	Input current, high level	VIII	X	0	20	0	20	0	20	0	20	0	20	µA
F6,F9,F14,F15,F22	Input current, high level	VIII	X	0	40	0	40	0	40	0	40	0	40	µA
F7,F16,F23	Input current, high level	VIII	X	0	60	0	60	0	60	0	60	0	60	µA
F8	Input current, high level	VIII	X	-248	-325	-248	-325	-248	-325	-248	-325	-248	-325	µA
F17	Input current, high level	VIII	X	0	80	0	80	0	80	0	80	0	80	µA
G1-G5,G10-G13,G18-G21	Input current, low level	IX	X	-10	-430	-10	-430	-10	-430	-10	-430	-10	-430	µA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
G6,G14,G15, G22	Input current, low level	IX	X	-10	-870	-10	-870	-10	-870	-10	-870	-10	-870	µA
G7,G9,G16,G23	Input current, low level	IX	X	-0.1	-1.3	-0.1	-1.3	-0.1	-1.3	-0.1	-1.3	-0.1	-1.3	mA
G8	Input current, low level	IX	X	-0.1	-1.8	-0.1	-1.8	-0.1	-1.8	-0.1	-1.8	-0.1	-1.8	mA
G17	Input current, low level	IX	X	-0.1	-2.6	-0.1	-2.6	-0.1	-2.6	-0.1	-2.6	-0.1	-2.6	mA
H1-H15	Output voltage, high level	X	X	2.5	4 5	2.5	4.5	2 5	4 5	2.5	4.5	2.5	4 5	V
J1-J15	Output voltage, low level	X	X	0	400	0	400	0	400	0	400	0	400	mV
K1-K42	Functional	XI	X	← See table XII →										-
L1	Power supply current, low level	XIII	X	10	102	10	102	10	102	10	102	10	102	mA
L2	Power supply current, high level	XIII	X	10	102	10	102	10	102	10	102	10	102	mA
M1-M12,M14, M15,M17-M22	Input capacitance	XIV	Q	2	20	2	20	2	20	2	20	2	20	pF
M13,M16	Input capacitance	XIV	Q	4	30	4	30	4	30	4	30	4	30	pF
M23	Input capacitance	XIV	Q	6	45	6	45	6	45	6	45	6	45	pF
N	Power supply transient amplitude	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
P1-P15	Transition time (t_{TLH}), C_L = 50 pF	XVI	S	10	50	10	60	10	60	10	70	10	70	ns
P16-P30	Transition time (t_{THL}), C_L = 50 pF	XVI	S	3	18	3	20	3	20	3	22	3	22	ns
Q1-Q15	Transition time (t_{TLH}), C_L = 150 pF	XVII	Q	20	80	20	90	20	90	20	100	20	100	ns
Q16-Q30	Transition time (t_{THL}), C_L = 150 pF	XVII	Q	4	40	4	45	4	45	4	50	4	50	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1-R4,R6-R9, R11-R14	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	2	24	2	26	2	26	2	28	2	28	ns
R5,R10,R15	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	3	39	4	41	4	41	4	43	4	43	ns
R16-R18	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	2	19	2	21	2	21	2	23	2	23	ns
R19-R24	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	2	29	3	31	3	31	3	33	3	33	ns
R25-R28,R30-R33,R35-R38, R40,R41,R42	Propagation delay time (t_{PHL}), C_L = 50 pF	XVIII	S	2	27	2	29	2	29	3	31	3	31	ns
R29,R34,R39	Propagation delay time (t_{PHL}), C_L = 50 pF	XVIII	S	3	39	4	41	4	41	4	43	4	43	ns
R43-R48	Propagation delay time (t_{PHL}), C_L = 50 pF	XVIII	S	3	33	3	35	3	35	3	37	3	37	ns
R49-R52,R54-R57,R59-R62	Propagation delay time (t_{PLH}) C_L = 150 pF	XVIII	Q	3	34	3	36	3	36	3	38	3	38	ns
R53,R58,R63	Propagation delay time (t_{PLH}), C_L = 150 pF	XVIII	Q	4	49	5	51	5	51	5	53	5	53	ns
R64-R66	Propagation delay time (t_{PLH}), C_L = 150 pF	XVIII	Q	2	29	3	31	3	31	3	33	3	33	ns
R67-R72	Propagation delay time (t_{PLH}), C_L = 150 pF	XVIII	Q	3	39	4	41	4	41	4	43	4	43	ns
R73-R76,R78-R81,R83-R86, R88-R90	Propagation delay time (t_{PHL}), C_L = 150 pF	XVIII	Q	3	37	3	39	3	39	4	41	4	41	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R77,R82,R87	Propagation delay time (t_{PHL}), C_L = 150 pF	XVIII	Q	4	49	5	51	5	51	5	53	5	53	ns
R91-R96	Propagation delay time (t_{PHL}), C_L = 150 pF	XVIII	Q	4	43	4	45	4	45	4	47	4	47	ns
S1-S3	Maximum clock frequency check	XIX	S	800	864	800	864	800	864	800	864	800	864	ns
T1-T12,T19-T24	Setup time (t_{SUH})	XX	S	850	1150	850	1150	850	1150	850	1150	850	1150	ns
T13-T18	Setup time (t_{SUH})	XX	S	1700	2300	1700	2300	1700	2300	1700	2300	1700	2300	ns
T25-T36,T43-T48	Setup time (t_{SUL})	XX	S	850	1150	850	1150	850	1150	850	1150	850	1150	ns
T37-T42	Setup time (t_{SUL})	XX	S	1700	2300	1700	2300	1700	2300	1700	2300	1700	2300	ns
U1-U12,U19-U24	Hold time (t_{HH})	XXI	S	850	1150	850	1150	850	1150	850	1150	850	1150	ns
U13-U18	Hold time (t_{HH})	XXI	S	1700	2300	1700	2300	1700	2300	1700	2300	1700	2300	ns
U25-U36,U43-U48	Hold time (t_{HL})	XXI	S	850	1150	850	1150	850	1150	850	1150	850	1150	ns
U37-U42	Hold time (t_{HL})	XXI	S	1700	2300	1700	2300	1700	2300	1700	2300	1700	2300	ns
V1-V15	Output crosstalk, high level	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
W1-W12	Output crosstalk, low level	XXIII	Q	0	600	0	600	0	600	0	600	0	600	mV
X	Power supply filter capacitance	XXIV	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	F

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, E, through L, and P through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through L and P through X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through X shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

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	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	561 max	561 max	mW
Input current (V_{IN} = 0.0 V)			
A,B,C,D,$\overline{ENP}$1	-430 max	-430 max	µA
U/$\overline{D}$1,U/$\overline{D}$2,U/$\overline{D}$3	-1.80 max	-1.80 max	mA
$\overline{ENT}$1,$\overline{ENT}$2,$\overline{ENT}$3,$\overline{ENP}$2,$\overline{ENP}$3	-870 max	-870 max	µA
LOAD1,LOAD2,LOAD3,CK1	-1.30 max	-1.30 max	mA
CK2,CK3	-2.60 max	-2.60 max	mA
Input current (V_{IN} = 2.7 V)			
A,B,C,D,$\overline{ENP}$1	20 max	20 max	µA
U/$\overline{D}$1,U/$\overline{D}$2,U/$\overline{D}$3	-325 max	-325 max	µA
$\overline{ENT}$1,$\overline{ENT}$2,$\overline{ENT}$3,$\overline{ENP}$2,$\overline{ENP}$3,CK1	40 max	40 max	µA
LOAD1,LOAD2,LOAD3	60 max	60 max	µA
CK2,CK3	80 max	80 max	µA
Output voltage			
Low level (I_{OL} = 4.0 mA max)	400 max	400 max	mV
High level (I_{OH} = -400 µA max)	2.5 max	2.5 max	V
Output current			
Low level (V_{OL} = 0.4 V max)	4.0 max	4.0 max	mA
High level (V_{OH} = 2.5 V max)	-400 max	-400 max	µA
Setup time			
A,B,C,D	20 min	20 min	ns
$\overline{ENP}$,$\overline{ENT}$	25 min	25 min	ns
LOAD	25 min	25 min	ns
U/$\overline{D}$	30 min	30 min	ns
Hold time (all inputs)	0 min	0 min	ns

* 6.5 Module function. The binary counter module, contains three 4 bit, synchronous binary, up/down counters. All three counters have a common up/down (U/$\overline{D}$) input and two have common clock (CK) and common parallel enable ($\overline{ENP}$) inputs. All other inputs and outputs are independent.

* 6.5.1 Input to output relationship. Each up/down counter module has the input to output relationship of table II for positive logic.

* 6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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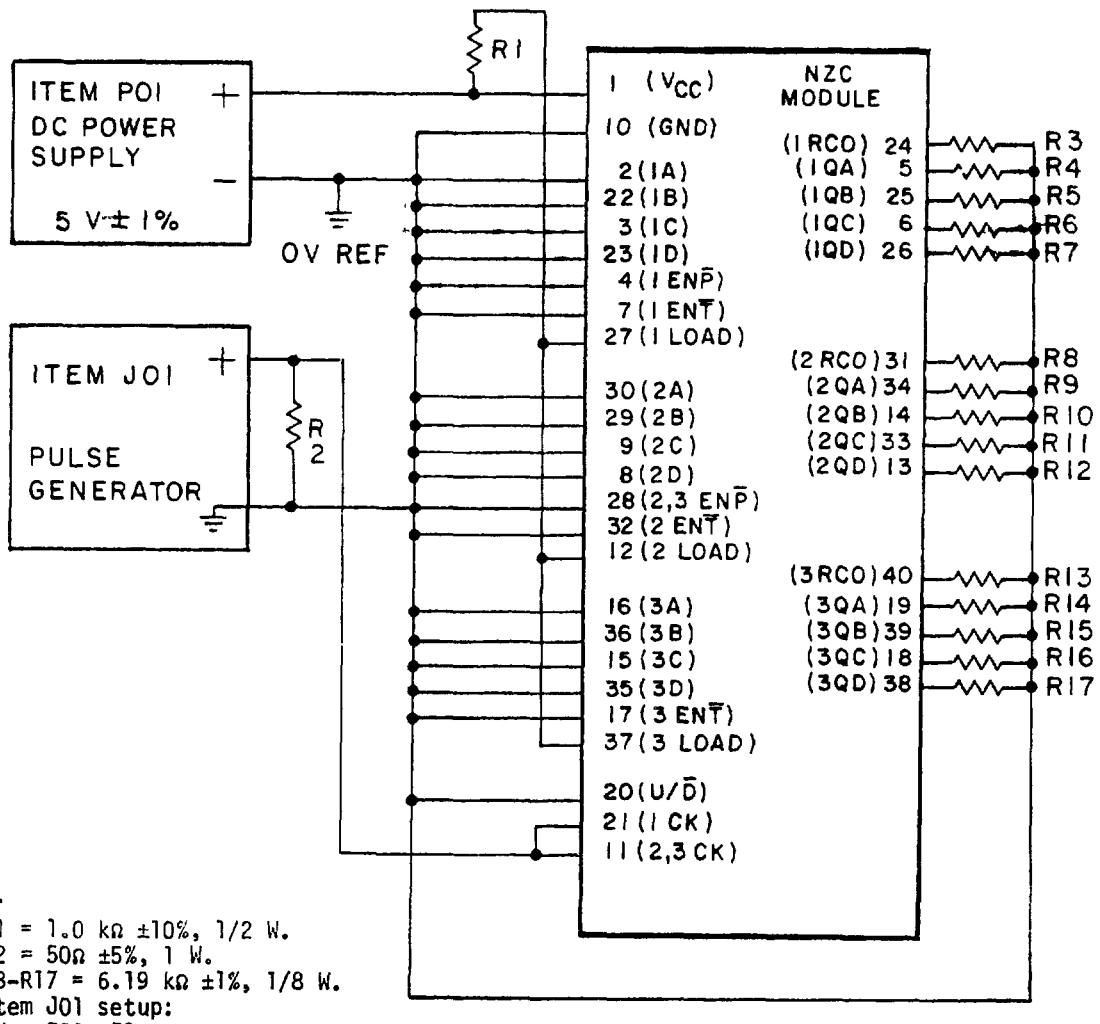
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TABLE II. Input to output relationship. 1/-5/

Inputs									Outputs					
$\overline{ENP}$	$\overline{ENT}$	U/$\overline{D}$	LOAD	CK	A	B	C	D	QA	QB	QC	QD	RCO	
L	L	L	L	↑	L	L	L	L	L	L	L	L	L	Load zeros
H	L	L	X	X	L	L	L	L	L	L	L	L	L	
X	H	L	X	X	L	L	L	L	L	L	L	L	H	
X	H	L	H	X	X	X	X	X	L	L	L	L	H	
H	L	L	H	X	X	X	X	X	L	L	L	L	L	
L	L	H	L	↑	H	H	H	H	H	H	H	H	L	Load ones
H	L	H	X	X	H	H	H	H	H	H	H	H	L	
X	H	H	X	X	H	H	H	H	H	H	H	H	H	
X	H	H	H	X	X	X	X	X	H	H	H	H	H	
H	L	H	H	X	X	X	X	X	H	H	H	H	L	
L	L	H	H	↑	X	X	X	X	L	L	L	L	H	0
L	L	H	H	↑	X	X	X	X						Count up on clock
L	L	H	H	↑	X	X	X	X	H	H	H	H	L	15
L	L	↓	H	C_L	X	X	X	X	H	H	H	H	↑	
L	L	L	H	↑	X	X	X	X	L	H	H	H	H	14
L	L	L	H	↑	X	X	X	X						Count down on clock
L	L	L	H	↑	X	X	X	X	L	L	L	L	L	0
L	L	L	H	↑	X	X	X	X	H	H	H	H	H	15

1/ H = High level (steady state).
2/ L = Low level (steady state).
3/ X = Don't care (including transition).
4/ ↑,↓ = Transition from low to high or high to low level.
5/ C_L = Clock level high or low before next transition.

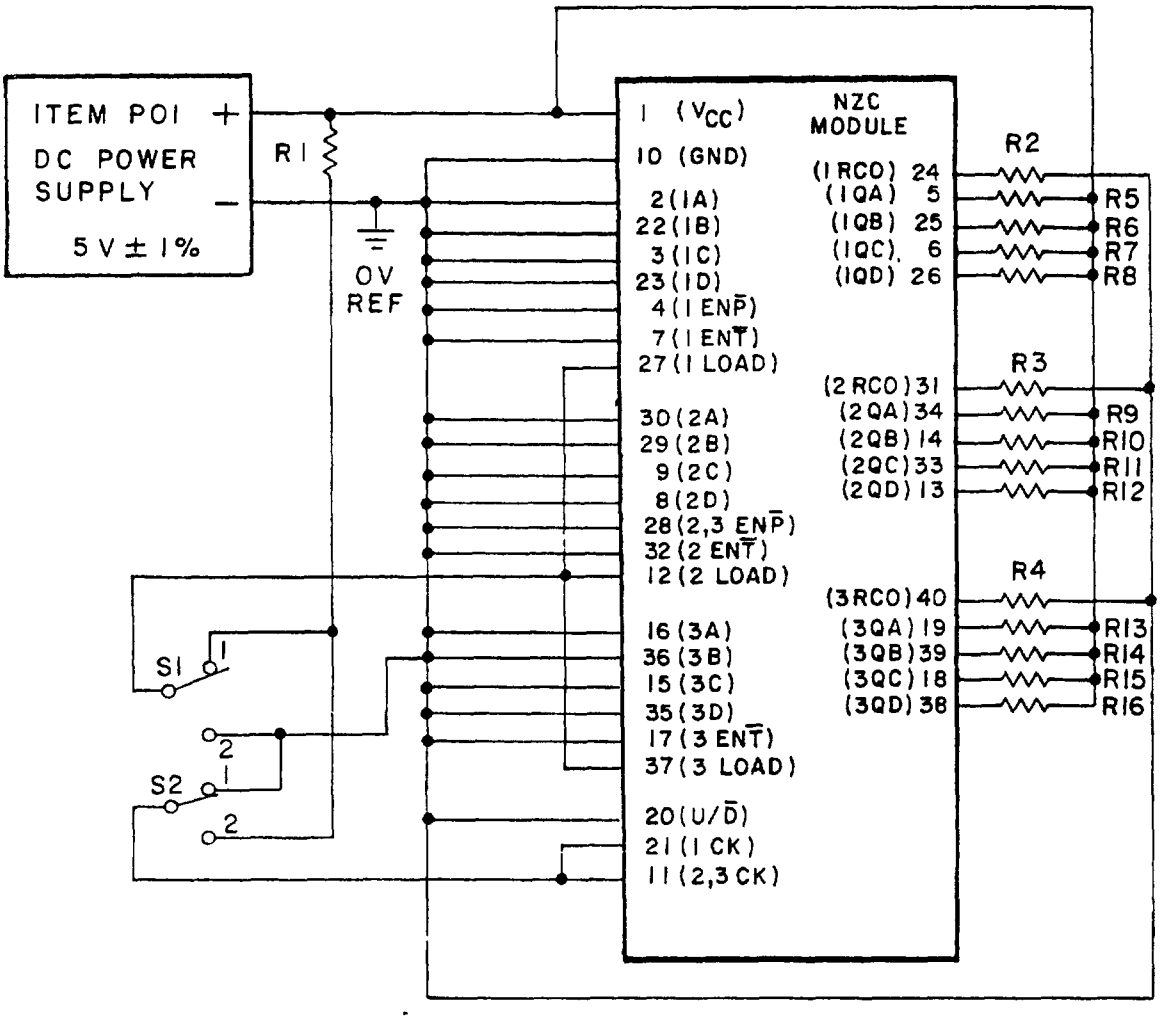
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NOTES:
1. R1 = 1.0 kΩ ±10%, 1/2 W.
2. R2 = 50Ω ±5%, 1 W.
3. R3-R17 = 6.19 kΩ ±1%, 1/8 W.
4. Item J01 setup:
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 6.0 ±2 ns.
   PA = 3.0 ±0.1 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 1,000 ±10%, 1/2 W.
2. R2-R4 = 6,190Ω ±1%, 1/8 W.
3. R5-R16 = 1,270Ω ±1%, 1/8 W.
4. S1= S2 = SPDT switch.
5. Switch S1 to position 2, then S2 to position 2 and then return S1 and S2 to position 1 to load zeros at start of test.

FIGURE 3. Component temperature test circuit.

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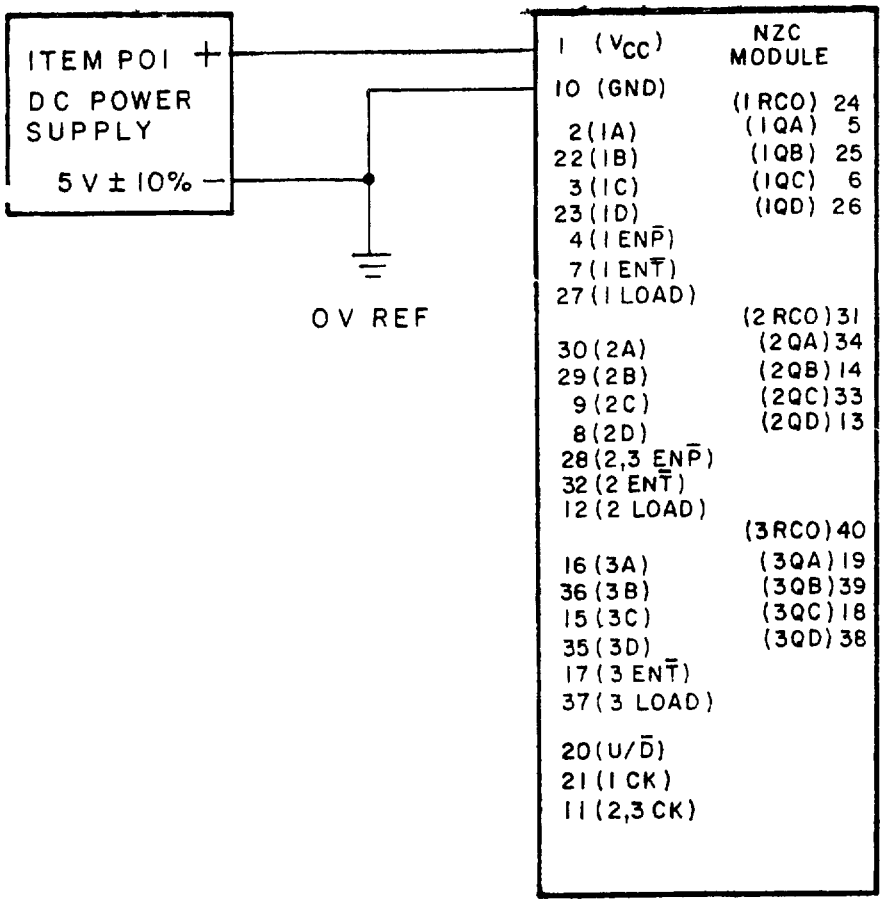


FIGURE 4. Vibration test circuit.

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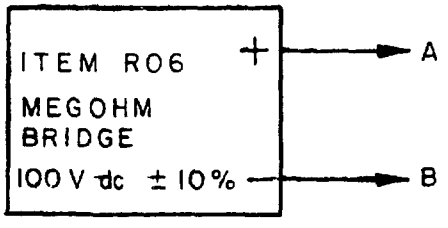
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	5

TABLE IV. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1RCO Output short circuit current	24	-15	-130	mA	6
B2	1QA	5				
B3	1QB	25				
B4	1QC	6				
B5	1QD	26				
B6	2RCO	31				
B7	2QA	34				
B8	2QB	14				
B9	2QC	33				
B10	2QD	13				
B11	3RCO	40				
B12	3QA	19				
B13	3QB	39				
B14	3QC	18				
B15	3QD	38				

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WARNING:

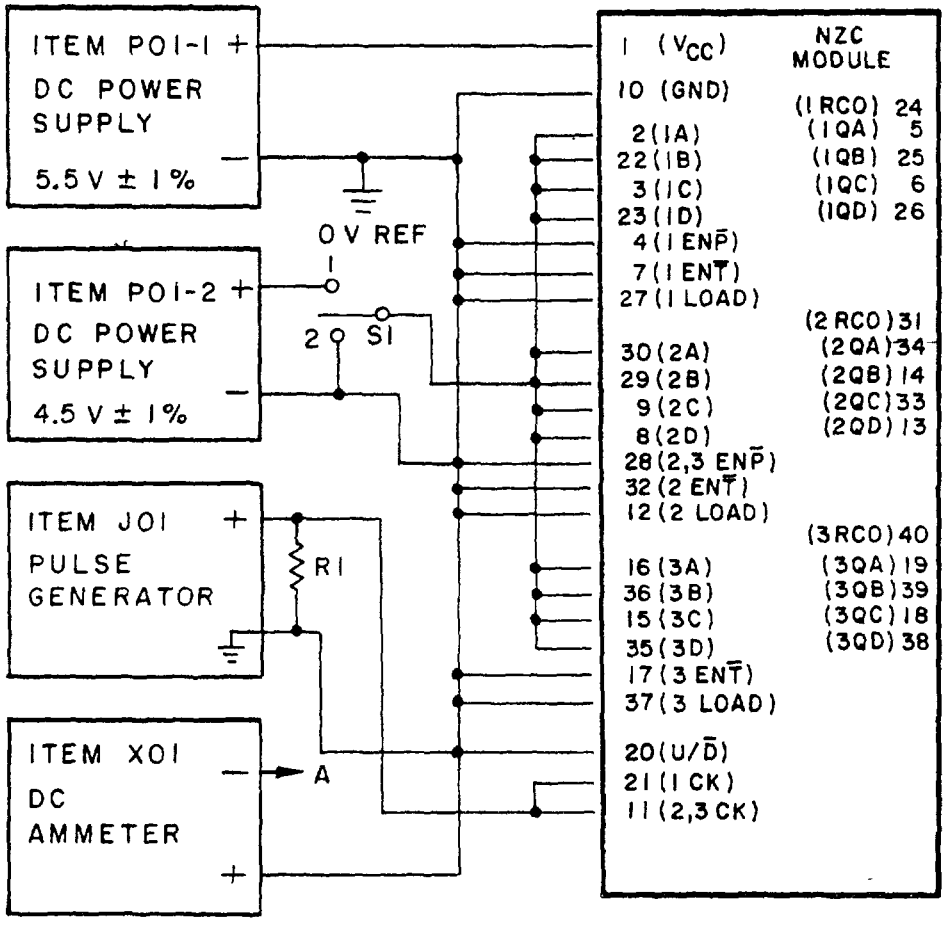
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

	NZC MODULE
1 (V_{CC})	
10 (GND)	
	(1RCO) 24
2(1A)	(1QA) 5
22(1B)	(1QB) 25
3(1C)	(1QC) 6
23(1D)	(1QD) 26
4(1 $\overline{ENP}$)	
7(1 $\overline{ENT}$)	
27(1 LOAD)	
	(2RCO)31
30(2A)	(2QA)34
29(2B)	(2QB)14
9(2C)	(2QC)33
8(2D)	(2QD)13
28(2,3 $\overline{ENP}$)	
32(2 $\overline{ENT}$)	
12(2 LOAD)	
	(3RCO)40
16(3A)	(3QA)19
36(3B)	(3QB)39
15(3C)	(3QC)18
35(3D)	(3QD)38
17(3 $\overline{ENT}$)	
37(3 LOAD)	
20(U/$\overline{D}$)	
21(1 CK)	
11(2,3 CK)	

FIGURE 5. Isolation test circuit for table III.

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Set item J01 to single pulse mode and refer to figure 12 for pulse parameters.
3. For test conditions B2-B5, B7-B10 and B12-B15, place S1 in position 1 and clock the module one time then take pins 7, 32, and 17 to position 1 of S1. For test conditions B1, B6 and B11 place S1 in position 2 and clock the module one time then take pins 7, 32, and 17 to position 1 of S1.
4. Only one output shall be shorted at a time and the measurements taken within 1 second.

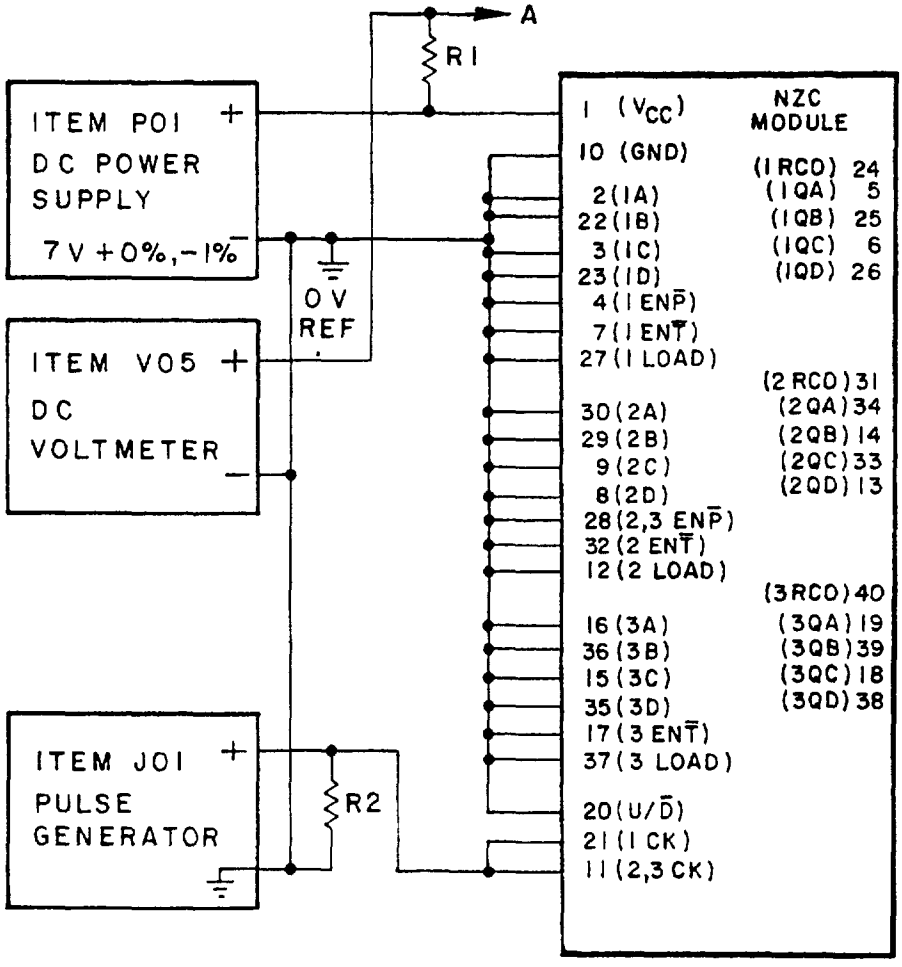
FIGURE 6. Output short circuit current test circuit for table IV.

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TABLE V. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	1RCO V_{OUT}, low loaded	24	10	800	mV	7
C2	1QA	5				
C3	1QB	25				
C4	1QC	6				
C5	1QD	26				
C6	2RCO	31				
C7	2QA	34				
C8	2QB	14				
C9	2QC	33				
C10	2QD	13				
C11	3RCO	40				
C12	3QA	19				
C13	3QB	39				
C14	3QC	18				
C15	3QD	38				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:43Z
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NOTES:
1. R1 = 1.27 kΩ ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.
3. Set item J01 to single pulse mode and refer to figure 12 for pulse parameters.
4. Pulse the module one time after all connections have been made.

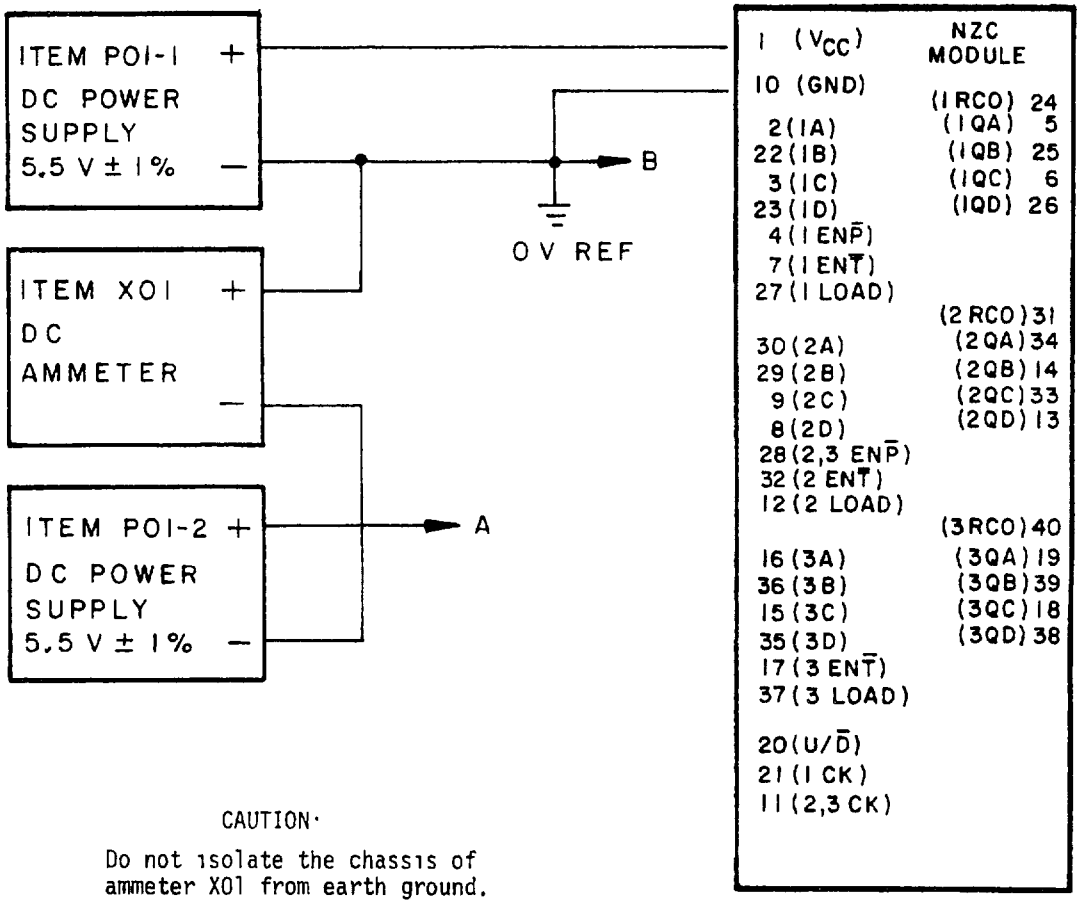
FIGURE 7. V_{OUT} (low-loaded) test circuit for table V.

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TABLE VI. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1A Input breakdown current	2	3,4,7,20,21,22,23,27	0	100	µA	8
D2	1B	22	2,3,4,7,20,21,23,27				
D3	1C	3	2,4,7,20,21,22,23,27				
D4	1D	23	2,3,4,7,20,21,22,27				
D5	1ENP̄	4	2,3,7,20,21,22,23,27				
D6	1ENT̄	7	2,3,4,20,21,22,23,27		200		
D7	1LOAD	27	2,3,4,7,20,21,22,23		300		
D8	U/D̄	20	2,3,4,7,21,22,23,27	-20	100		
D9	1CK	21	2,3,4,7,20,22,23,27	0	200		
D10	2A	30	8,9,11,12,28,29,32		100		
D11	2B	29	8,9,11,12,28,30,32				
D12	2C	9	8,11,12,28,29,30,32				
D13	2D	8	9,11,12,28,29,30,32				
D14	2,3ENP̄	28	8,9,11,12,29,30,32		200		
D15	2ENT̄	32	8,9,11,12,28,29,30		200		
D16	2LOAD	12	8,9,11,28,29,30,32		300		
D17	2,3CK	11	8,9,12,28,29,30,32		400		
D18	3A	16	11,15,17,35,36,37		100		
D19	3B	36	11,15,16,17,35,37				
D20	3C	15	11,16,17,35,36,37				
D21	3D	35	11,15,16,17,36,37				
D22	3ENT̄	17	11,15,16,35,36,37		200		
D23	3LOAD	37	11,15,16,17,35,36		300		

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CAUTION:
Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: Power supply PO1-2 shall be current limited to 5.0 mA.

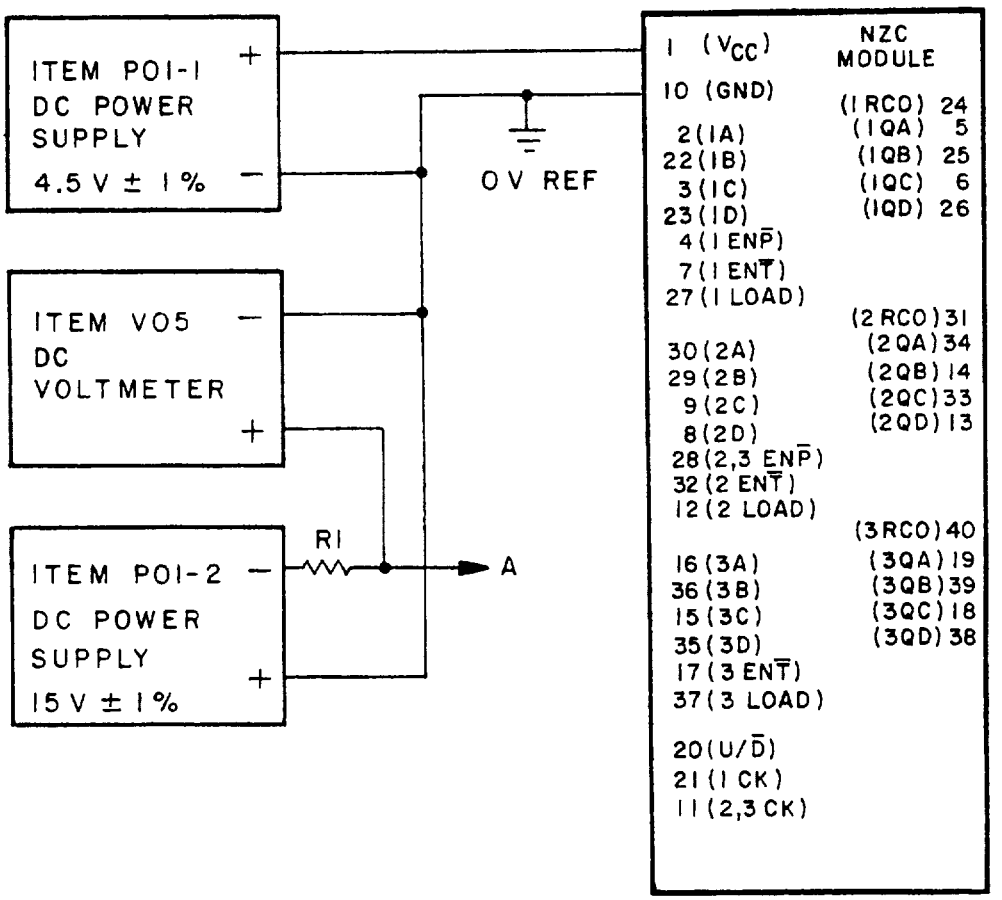
FIGURE 8. Input breakdown current test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
E1	U/$\overline{D}$ Input clamp diode voltage	20	-0.4	-1.5	V	9
E2	1CK	21				
E3	1A	2				
E4	1B	22				
E5	1C	3				
E6	1D	23				
E7	1EN$\overline{P}$	4				
E8	1LOAD	27				
E9	1EN$\overline{T}$	7				
E10	2,3CK	11				
E11	2A	30				
E12	2B	29				
E13	2C	9				
E14	2D	8				
E15	2,3EN$\overline{P}$	28				
E16	2LOAD	12				
E17	2EN$\overline{T}$	32				
E18	3A	16				
E19	3B	36				
E20	3C	15				
E21	3D	35				
E22	3LOAD	37				
E23	3EN$\overline{T}$	17				

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NOTE: R1 = 750Ω ±1%, 1/2 W.

* FIGURE 9. Input clamp diode voltage test circuit for table VII.

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TABLE VIII. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1A Input current, high level	2	Open	1/	0	20	µA	10
F2	1B	22						
F3	1C	3						
F4	1D	23						
F5	1ENP̄	4						
F6	1ENT̄	7				40		
F7	1LOAD	27				60		
F8	U/D̄	20			-248	-325		
F9	1CK	21			0	40		
F10	2A	30				20		
F11	2B	29						
F12	2C	9						
F13	2D	8						
F14	2,3ENP̄	28				40		
F15	2ENT̄	32				40		
F16	2LOAD	12				60		
F17	2,3CK	11				80		
F18	3A	16				20		
F19	3B	36						
F20	3C	15						
F21	3D	35						
F22	3ENT̄	17				40		
F23	3LOAD	37				60		

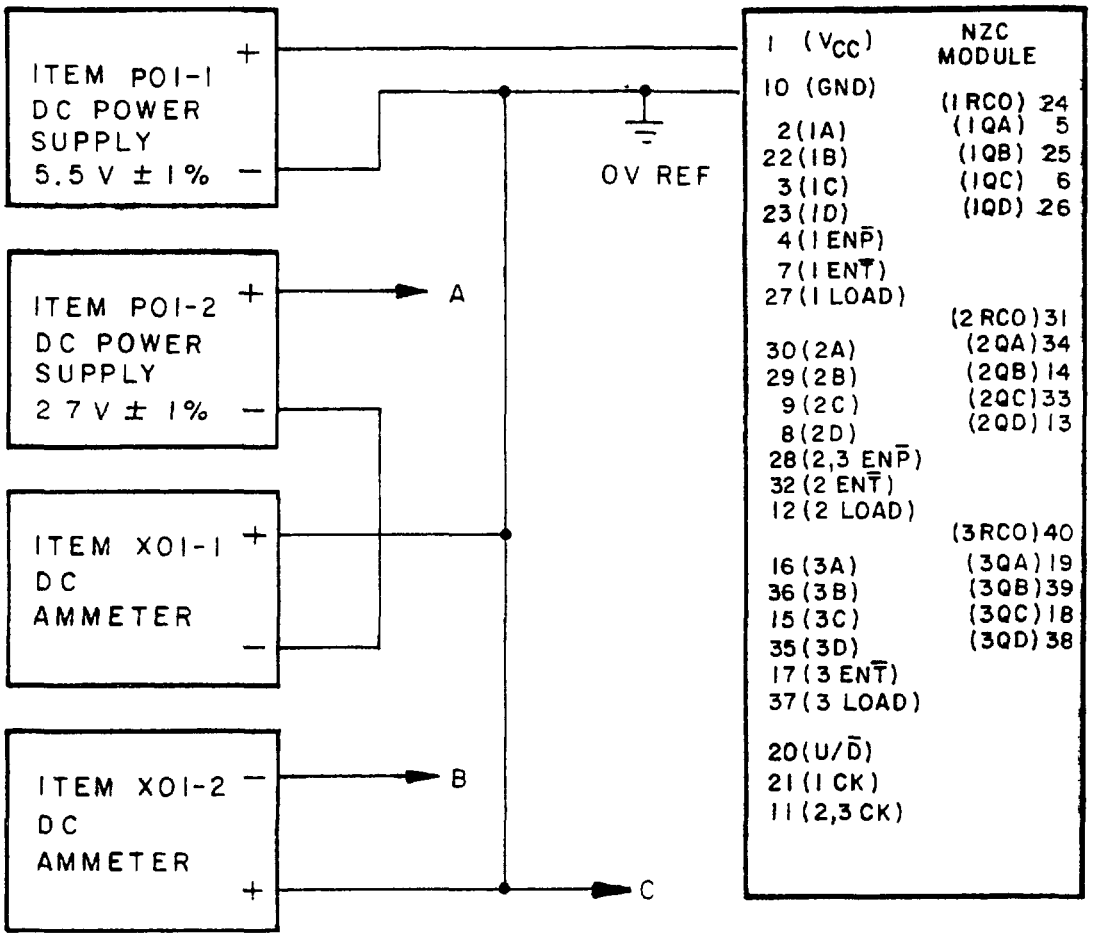
1/ All input pins except the one connected to pin A shall be connected to pin C.

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* TABLE IX. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1A Input current, low level	Open	2	Open	-10	-430	µA	10
G2	1B		22					
G3	1C		3					
G4	1D		23					
G5	1ENP		4					
G6	1ENT		7			-870		
G7	1LOAD		27		-0.1	-1.3	mA	
G8	U/D		20		-0.1	-1.8	mA	
G9	1CK		21		-0.1	-1.3	mA	
G10	2A		30		-10	-430	µA	
G11	2B		29					
G12	2C		9					
G13	2D		8					
G14	2ENP		28			-870		
G15	2ENT		32			-870		
G16	2LOAD		12		-0.1	-1.3	mA	
G17	2,3CK		11		-0.1	-2.6	mA	
G18	3A		16		-10	-430	µA	
G19	3B		36					
G20	3C		15					
G21	3D		35					
G22	3ENT		17			-870		
G23	3LOAD		37		-0.1	-1.3	mA	

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Since ammeters X01-1 and X01-2 are not used simultaneously, only one ammeter X01 is required.

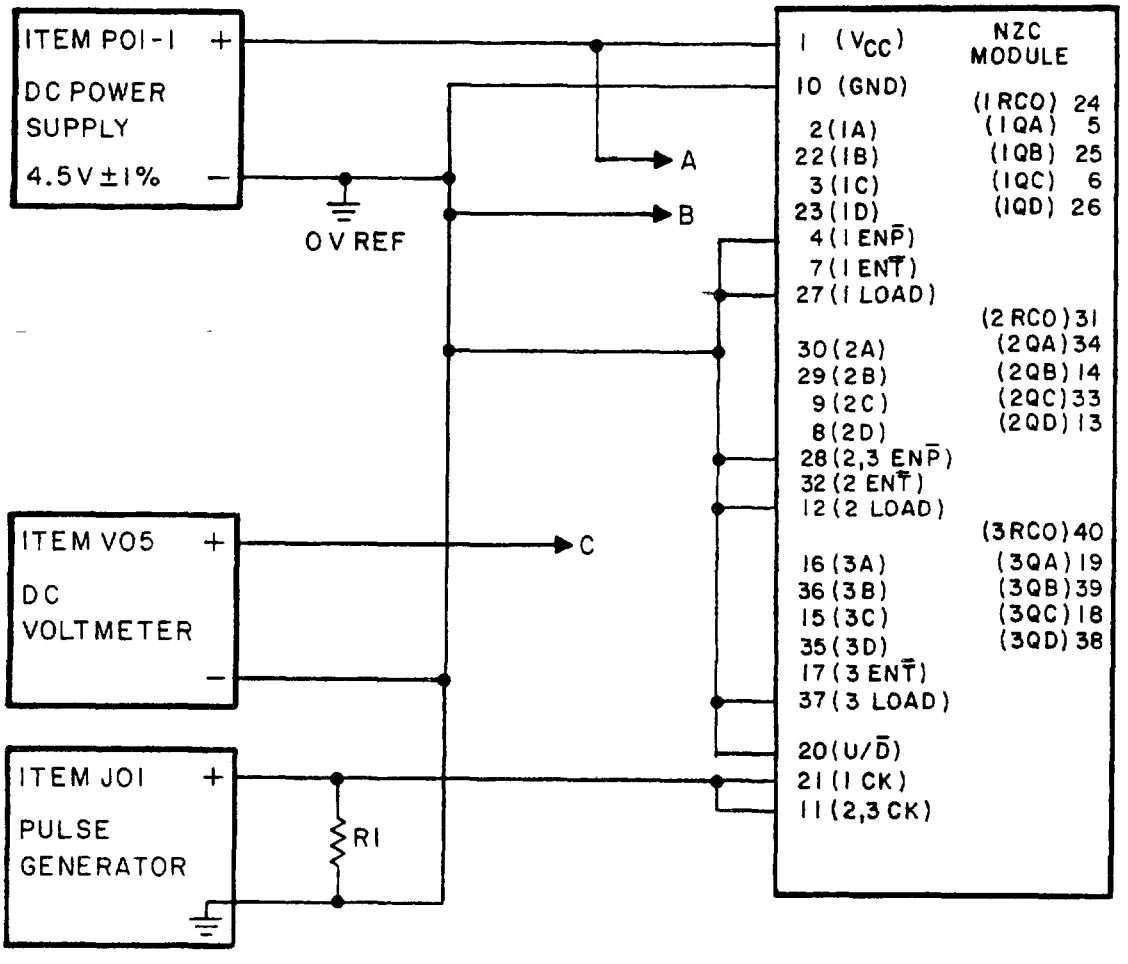
FIGURE 10. Input current (high-level and low-level) test circuit for tables VIII and IX.

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* TABLE X. Output voltage (high-level and low-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	1QA	Output voltage, high level	2	3,7,22,23	5	2.5	4.5	V	11
H2	1QB		22	2,3,7,23	25				
H3	1QC		3	2,7,22,23	6				
H4	1QD		23	2,3,7,22	26				
H5	1RCO		7	2,3,22,23	24				
H6	2QA		30	8,9,29,32	34				
H7	2QB		29	8,9,30,32	14				
H8	2QC		9	8,29,30,32	33				
H9	2QD		8	9,29,30,32	13				
H10	2RCO		32	8,9,29,30	31				
H11	3QA		16	15,17,35,36	19				
H12	3QB		36	15,16,17,35	39				
H13	3QC		15	16,17,35,36	18				
H14	3QD		35	15,16,17,36	38				
H15	3RCO		17	15,16,35,36	40				
J1	1QA	Output voltage, low level	Open	2,3,7,22,23	5	0	400	mV	
J2	1QB				25				
J3	1QC				6				
J4	1QD				26				
J5	1RCO				24				
J6	2QA			8,9,29,30,32	34				
J7	2QB				14				
J8	2QC				33				
J9	2QD				13				
J10	2RCO				31				
J11	3QA			15,16,17,35,36	19				
J12	3QB				39				
J13	3QC				18				
J14	3QD				38				
J15	3RCO				40				

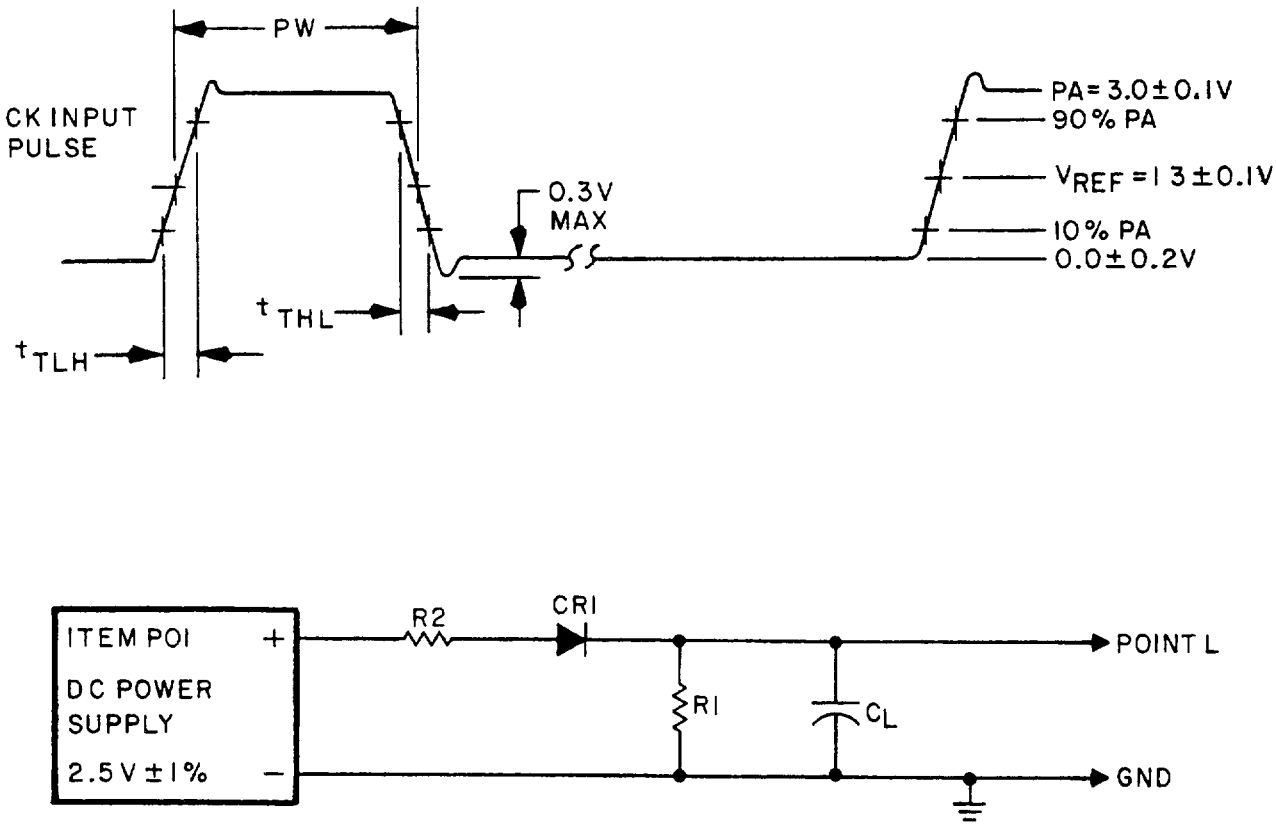
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Set item J01 to single pulse mode and load outputs as shown on figure 12.
3. Pulse the module one time for each test condition.

FIGURE 11. Output voltage (high-level and low-level) test circuit for table X.

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NOTES:
1. Clock input pulse parameters:
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 6.0 \pm 2$ ns.
2. CR1 = 1N3600, 1N4150 or equivalent.
3. R1 = 6.19 kΩ ±1%, 1/8 W.
4. R2 = Select value for a current flow of 4.0 mA ±1% out of point L with point L held at 0.4 V.
5. C_L is not required for dc electrical tests H and J.
6. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions P1-P30, R1-R48, S1-S3, T1-T48, U1-U48, V1-V15 and W1-W12.
7. C_L = 150 ±5 pF (including probe and parasitic capacitance) for test conditions Q1-Q30 and R49-R96.
8. Lead length from load circuit to module output pin under test shall be less than 2 inches (50.8 mm).

FIGURE 12. Clock-input pulse and logic load circuit.

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TABLE XI. Functional test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	Functional	7,17,32	2-4,8,9,12,15,16,20,22,23,27-30,35-37	See table XII	See note 4, figure 13		--	13,14
K2		2,3,7-9,15-17,22,23,29,30,32,35,36	4,12,20,27,28,37					
K3		Open	2-4,7-9,12,15-17,20,22,23,27-30,32,35-37					
K4		2,3,8,9,15,16,22,23,29,30,35,36	4,7,12,17,20,27,28,32,37					
K5		20	2-4,7-9,12,15-17,22,23,27-30,32,35-37					
K6		2,3,8,9,15,16,20,22,23,29,30,35,36	4,7,12,17,27,28,32,37					
K7		7,12,17,27,32,37	2-4,8,9,15,16,20,22,23,28-30,35,36					
K8		4,7,12,17,27,28,32,37	2,3,8,9,15,16,20,22,23,29,30,35,36					
K9		4,12,27,28,37	2,3,7-9,15-17,20,22,23,29,30,32,35,36					
K10	Functional, count up	12,20,27,37	2-4,7-9,15-17,22,23,28-30,32,35,36					
K11								
K12								
K13								
K14								
K15								
K16								

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Check the source to verify that this is the current version before use.

TABLE XI. Functional test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K17	Functional, count up	12,20,27,37	2-4,7-9,15-17,22,23,28-30,32,35,36	See table XII	See note 4, figure 13		--	13,14
K18								
K19								
K20								
K21								
K22								
K23								
K24								
K25								
K26								
K27	Functional, count down	12,27,37	2-4,7-9,15-17,20,22,23,28-30,32,35,36					
K28								
K29								
K30								
K31								
K32								
K33								
K34								
K35								
K36								
K37								
K38								
K39								
K40								
K41								
K42								

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TABLE XII. Output status.

Test condition	Output pins																								
	24	26	6	25	5	31	13	33	14	34	40	38	18	39	19										
K1	H	L	L	L	L	H	L	L	L	L	H	L	L	L	L										
K2	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H										
K3	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L										
K4	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H										
K5	H	L	L	L	L	H	L	L	L	L	H	L	L	L	L										
K6	L	H	H	H	H	L	H	H	H	H	L	H	H	H	H										
K7	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H										
K8	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H										
K9	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H										
K10	H	L	L	L	L	H	L	L	L	L	H	L	L	L	L										
K11	H	L	L	L	H	H	L	L	L	H	H	L	L	L	H										
K12	H	L	L	H	L	H	L	L	H	L	H	L	L	H	L										
K13	H	L	L	H	H	H	L	L	H	H	H	L	L	H	H										
K14	H	L	H	L	L	H	L	H	L	L	H	L	H	L	L										
K15	H	L	H	L	H	H	L	H	L	H	H	L	H	L	H										
K16	H	L	H	H	L	H	H	H	H	L	H	L	H	H	L										
K17	H	L	H	H	H	H	H	H	H	H	H	L	H	H	H										
K18	H	H	L	L	L	H	H	L	L	L	H	H	L	L	L										
K19	H	H	L	L	H	H	H	L	L	H	H	H	L	L	H										
K20	H	H	L	H	L	H	H	L	H	L	H	H	L	H	L										
K21	H	H	L	H	H	H	H	L	H	H	H	H	L	H	H										
K22	H	H	H	L	L	H	H	H	L	L	H	H	H	L	L										
K23	H	H	H	L	H	H	H	H	L	H	H	H	H	L	H										
K24	H	H	H	H	L	H	H	H	H	L	H	H	H	H	L										
K25	L	H	H	H	H	L	H	H	H	H	L	H	H	H	H										

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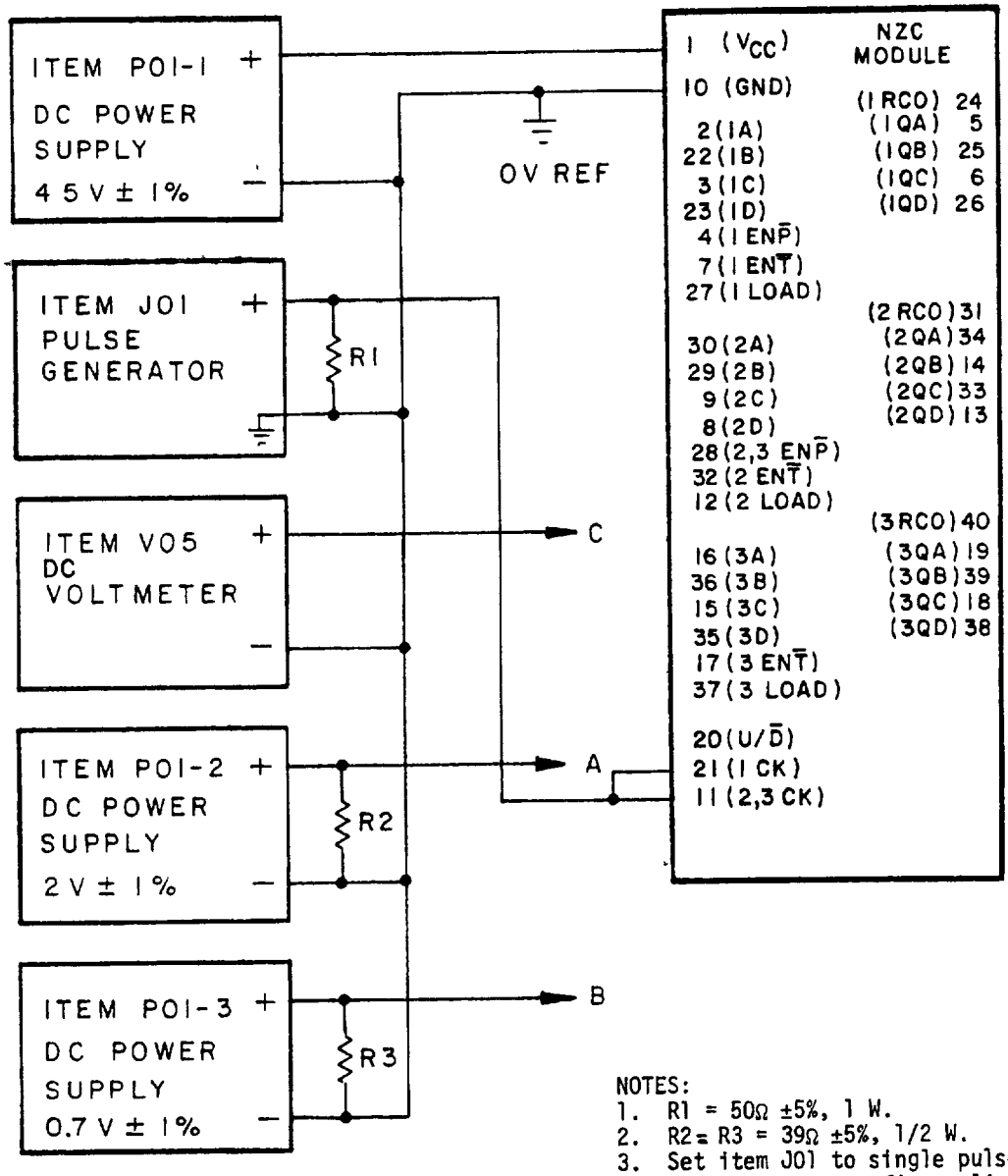
TABLE XII. Output status - Continued.

Test condition	Output pins														
	24	26	6	25	5	31	13	33	14	34	40	38	18	39	19
K26	H	L	L	L	L	H	L	L	L	L	H	L	L	L	L
K27	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
K28	H	H	H	H	L	H	H	H	H	L	H	H	H	H	L
K29	H	H	H	L	H	H	H	H	L	H	H	H	H	L	H
K30	H	H	H	L	L	H	H	H	L	L	H	H	H	L	L
K31	H	H	L	H	H	H	H	L	H	H	H	H	L	H	H
K32	H	H	L	H	L	H	H	L	H	L	H	H	L	H	L
K33	H	H	L	L	H	H	H	L	L	H	H	H	L	L	H
K34	H	H	L	L	L	H	H	L	L	L	H	H	L	L	L
K35	H	L	H	H	H	H	L	H	H	H	H	L	H	H	H
K36	H	L	H	H	L	H	L	H	H	L	H	L	H	H	L
K37	H	L	H	L	H	H	L	H	L	H	H	L	H	L	H
K38	H	L	H	L	L	H	L	H	L	L	H	L	H	L	L
K39	H	L	L	H	H	H	L	L	H	H	H	L	L	H	H
K40	H	L	L	H	L	H	L	L	H	L	H	L	L	H	L
K41	H	L	L	L	H	H	L	L	L	H	H	L	L	L	H
K42	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L

TABLE XIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L1	Power supply current, low level	10	102	mA	15
L2	Power supply current, high level	10	102	mA	15

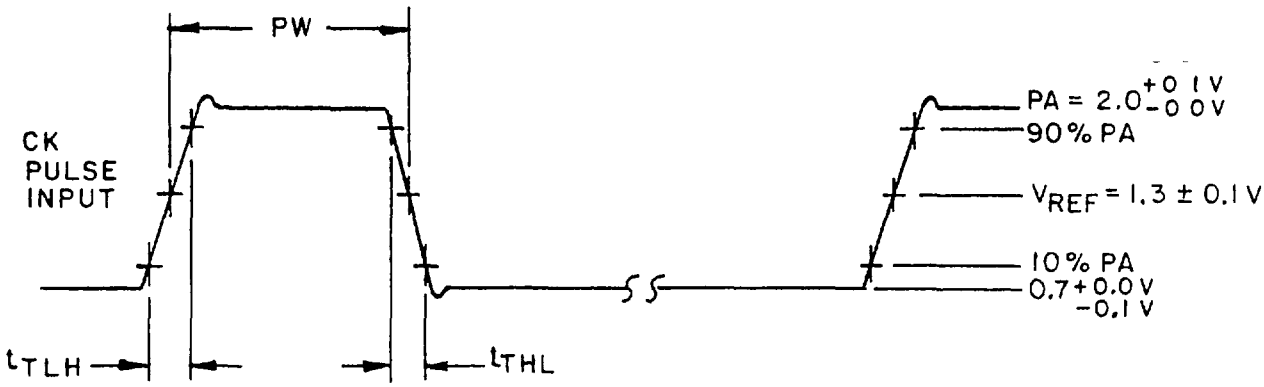
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = R3 = 39Ω ±5%, 1/2 W.
3. Set item J01 to single pulse mode as shown on figure 14.
4. Verify that the output low level is ≤ 0.4 V and the output high level is ≥ 2.5 V by connecting item V05 (point C) to each output pin after input conditions and clock pulse have been applied.

* FIGURE 13. Functional test circuit for tables XI and XII.

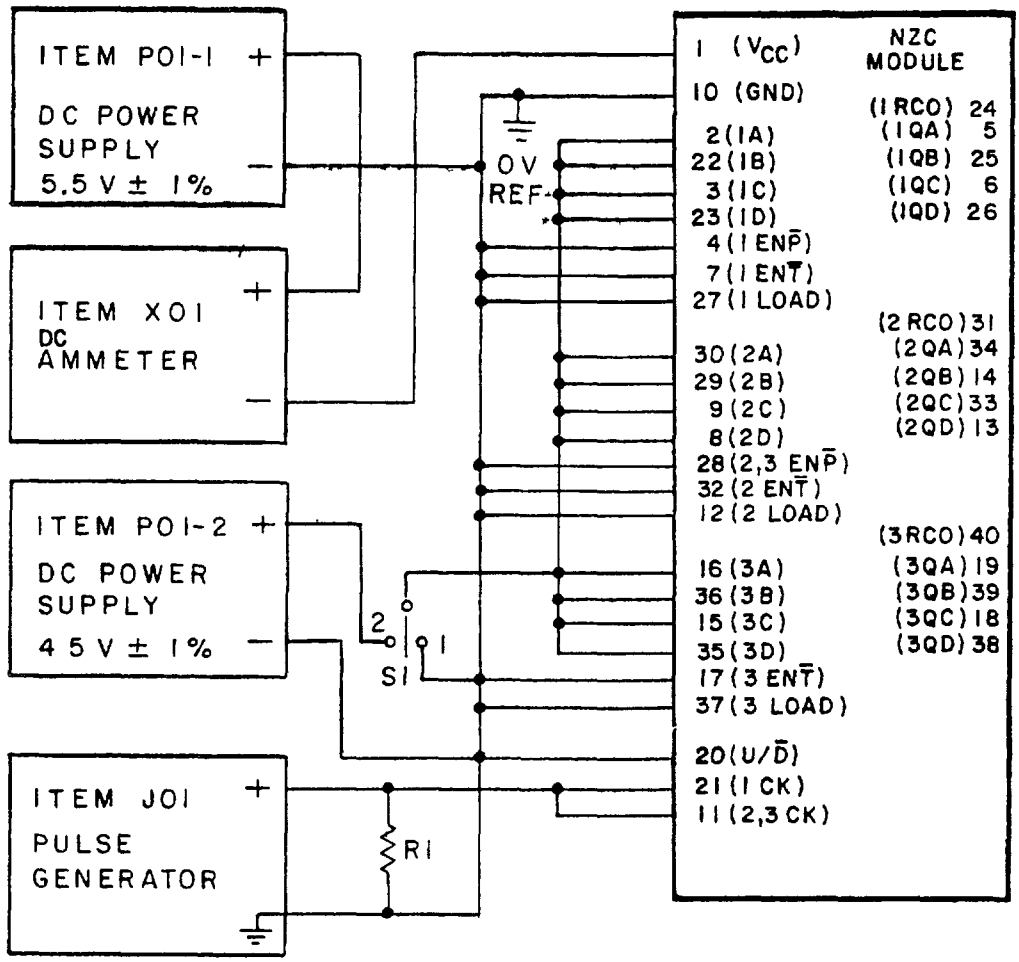
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NOTE: Input pulse parameters:
PW = 500 ±50 ns.
$t_{TLH} = t_{THL} = 6.0 \pm 2$ ns.

FIGURE 14. Functional test circuit waveform for tables XI and XII.

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. S1 = SPDT switch.
3. Set item J01 to generate a single pulse and adjust waveform as shown on figure 12.
4. Gate a single pulse to the module under test after setting S1 to position 1 for test condition L1 and position 2 for test condition L2.

*FIGURE 15. Power supply current (low-level and high-level) test circuit for table XIII.

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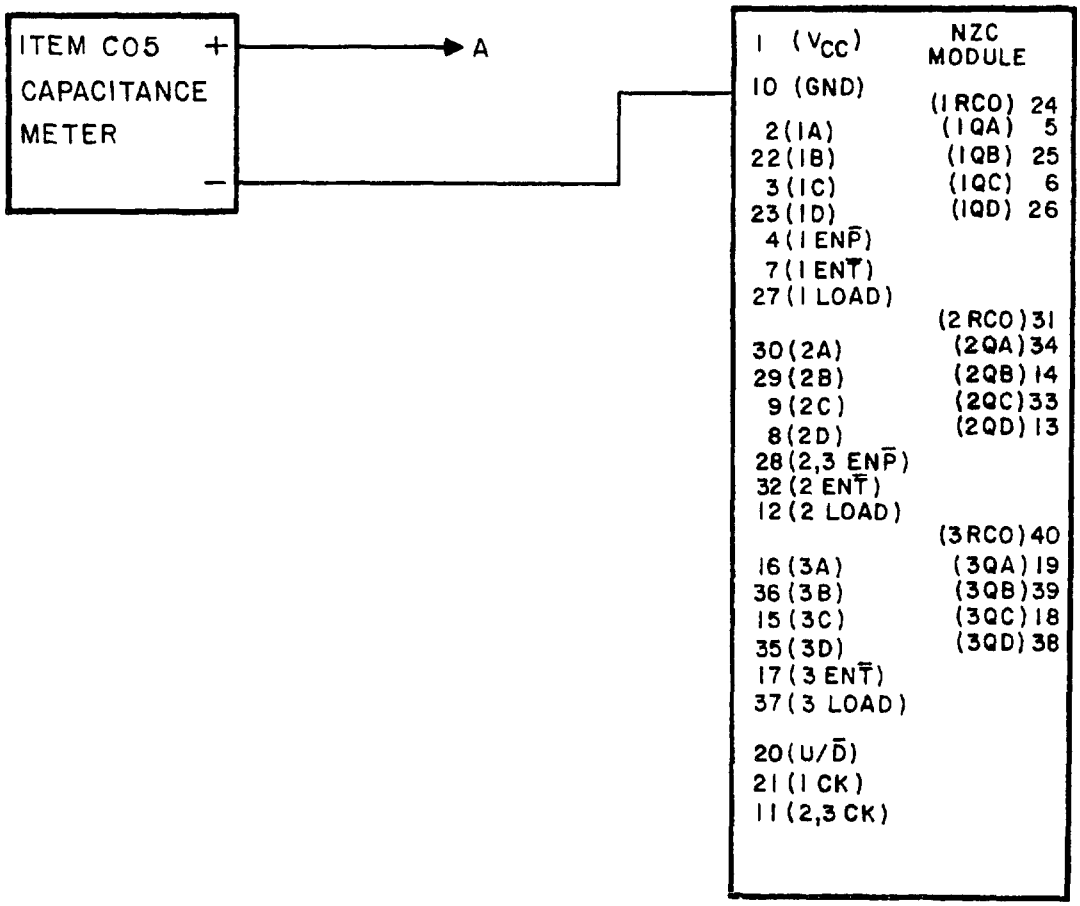
* TABLE XIV. Input capacitance test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
M1	1A Input capacitance	2	2	20	pF	16
M2	1B	22				
M3	1C	3				
M4	1D	23				
M5	1ENP̄	4				
M6	1ENT̄	7				
M7	1LOAD̄	27				
M8	1CK	21				
M9	2A	30				
M10	2B	29				
M11	2C	9				
M12	2D	8				
M13	2,3ENP̄	28	4	30		
M14	2ENT̄	32	2	20		
M15	2LOAD	12	2	20		
M16	2,3CK	11	4	30		
M17	3A	16	2	20		
M18	3B	36				
M19	3C	15				
M20	3D	35				
M21	3ENT̄	17				
M22	3LOAD	37				
M23	U/D̄	20	6	45		

TABLE XV. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
N	Power supply transient amplitude	0	400	mV	17,18

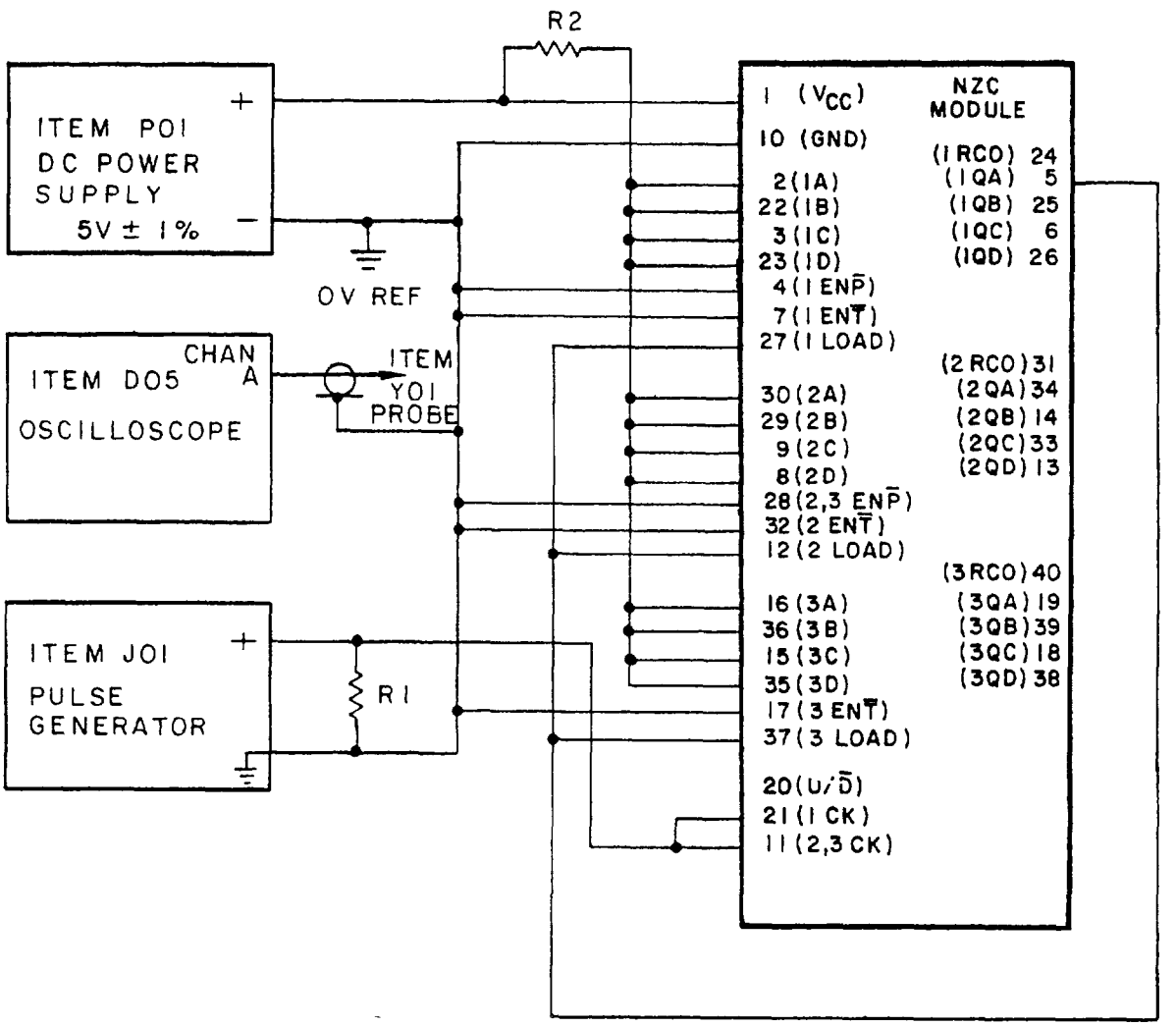
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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 16. Input capacitance test circuit for table XIV.

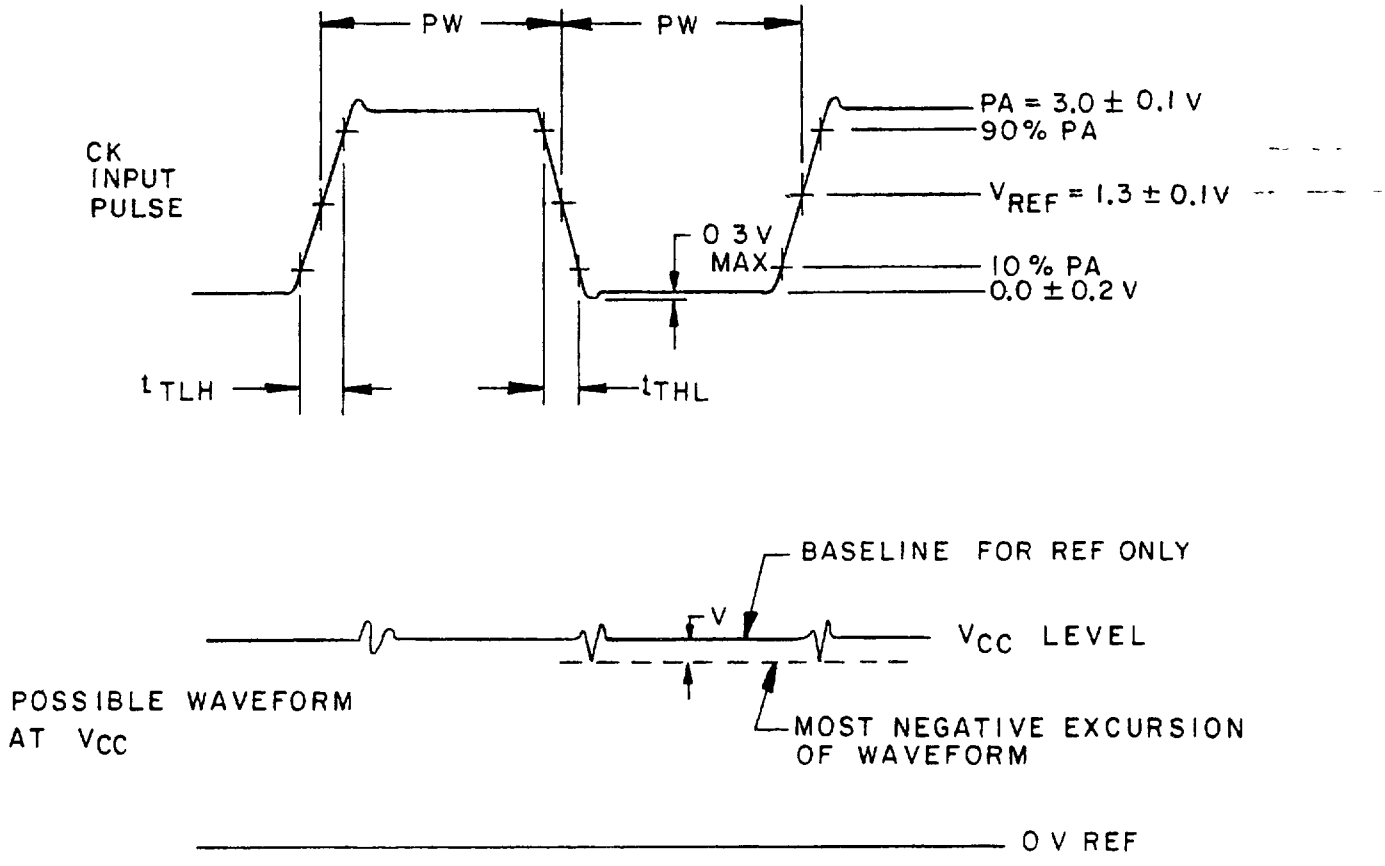
Source: https://assist.dla.mil -- Downloaded: 20260-20T17:43Z
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 1 kΩ ±10%, 1/2 W.
3. Set item J01 to generate clock pulses as shown on figure 18.

FIGURE 17. Power supply transient amplitude test circuit for table XV.

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NOTES:
1. Input pulse parameters.
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 6.0 \pm 2$ ns.
2. V = Measurement is made from the base line to the most negative excursion of the waveform.

FIGURE 18. Power supply transient amplitude test circuit waveforms for table XV.

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TABLE XVI. Transition time (t_{TLH} and t_{THL}) test.

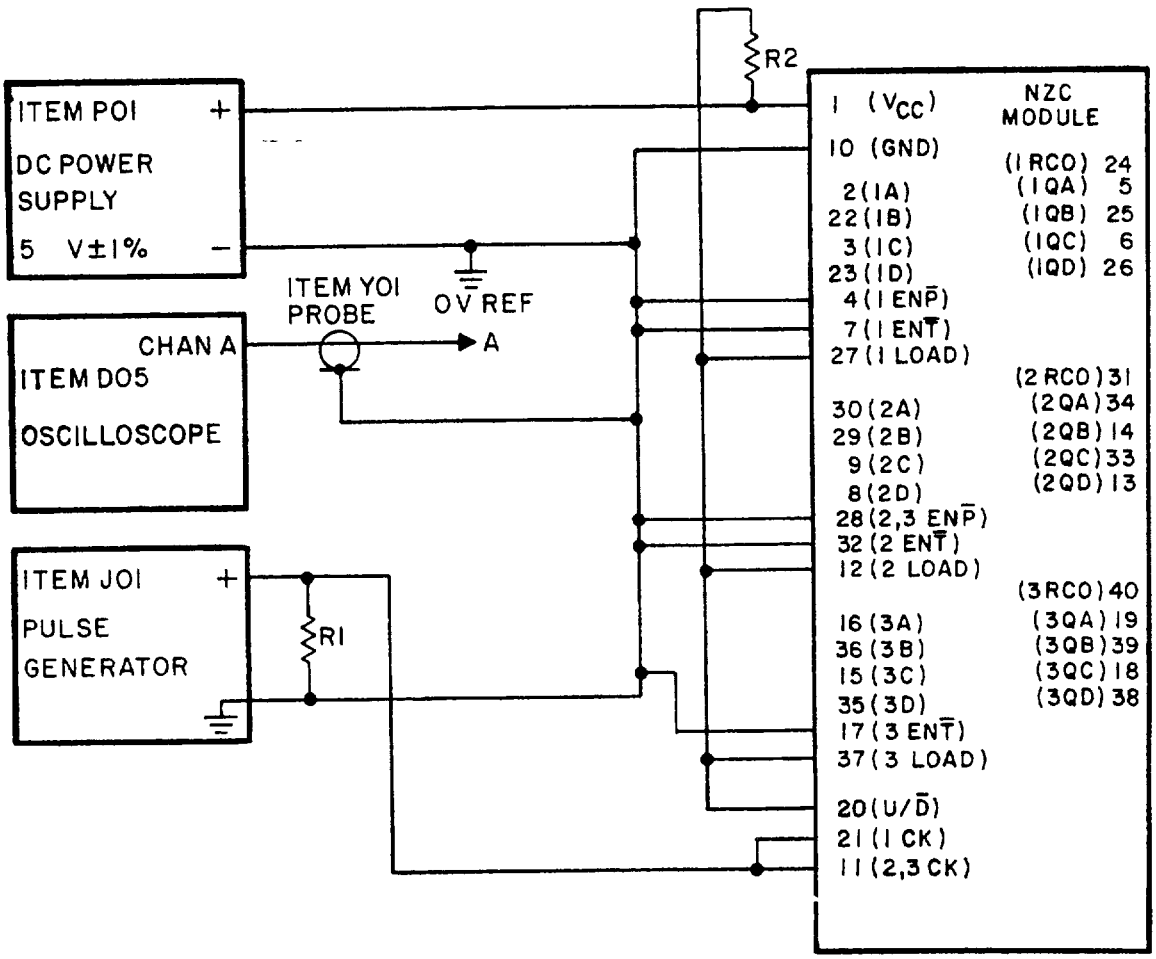
Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
P1	1QA	Transition time (t_{TLH}), C_L = 50 pF	5	10	50	ns	19,20
P2	1QB		25				
P3	1QC		6				
P4	1QD		26				
P5	1RCO		24				
P6	2QA		34				
P7	2QB		14				
P8	2QC		33				
P9	2QD		13				
P10	2RCO		31				
P11	3QA		19				
P12	3QB		39				
P13	3QC		18				
P14	3QD		38				
P15	3RCO		40				
P16	1QA	Transition time (t_{THL}), C_L = 50 pF	5	3	18		
P17	1QB		25				
P18	1QC		6				
P19	1QD		26				
P20	1RCO		24				
P21	2QA		3				
P22	2QB		14				
P23	2QC		33				
P24	2QD		13				
P25	2RCO		31				
P26	3QA		19				
P27	3QB		39				
P28	3QC		18				
P29	3QD		38				
P30	3RCO		40				

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* TABLE XVII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
Q1	1QA	Transition time (t_{TLH}), C_L = 150 pF	5	20	80	ns	19,20
Q2	1QB		25				
Q3	1QC		6				
Q4	1QD		26				
Q5	1RCO		24				
Q6	2QA		34				
Q7	2QB		14				
Q8	2QC		33				
Q9	2QD		13				
Q10	2RCO		31				
Q11	3QA		19				
Q12	3QB		39				
Q13	3QC		18				
Q14	3QD		38				
Q15	3RCO		40				
Q16	1QA	Transition time (t_{THL}), C_L = 150 pF	5	4	40		
Q17	1QB		25				
Q18	1QC		6				
Q19	1QD		26				
Q20	1RCO		24				
Q21	2QA		34				
Q22	2QB		14				
Q23	2QC		33				
Q24	2QD		13				
Q25	2RCO		31				
Q26	3QA		19				
Q27	3QB		39				
Q28	3QC		18				
Q29	3QD		38				
Q30	3RCO		40				

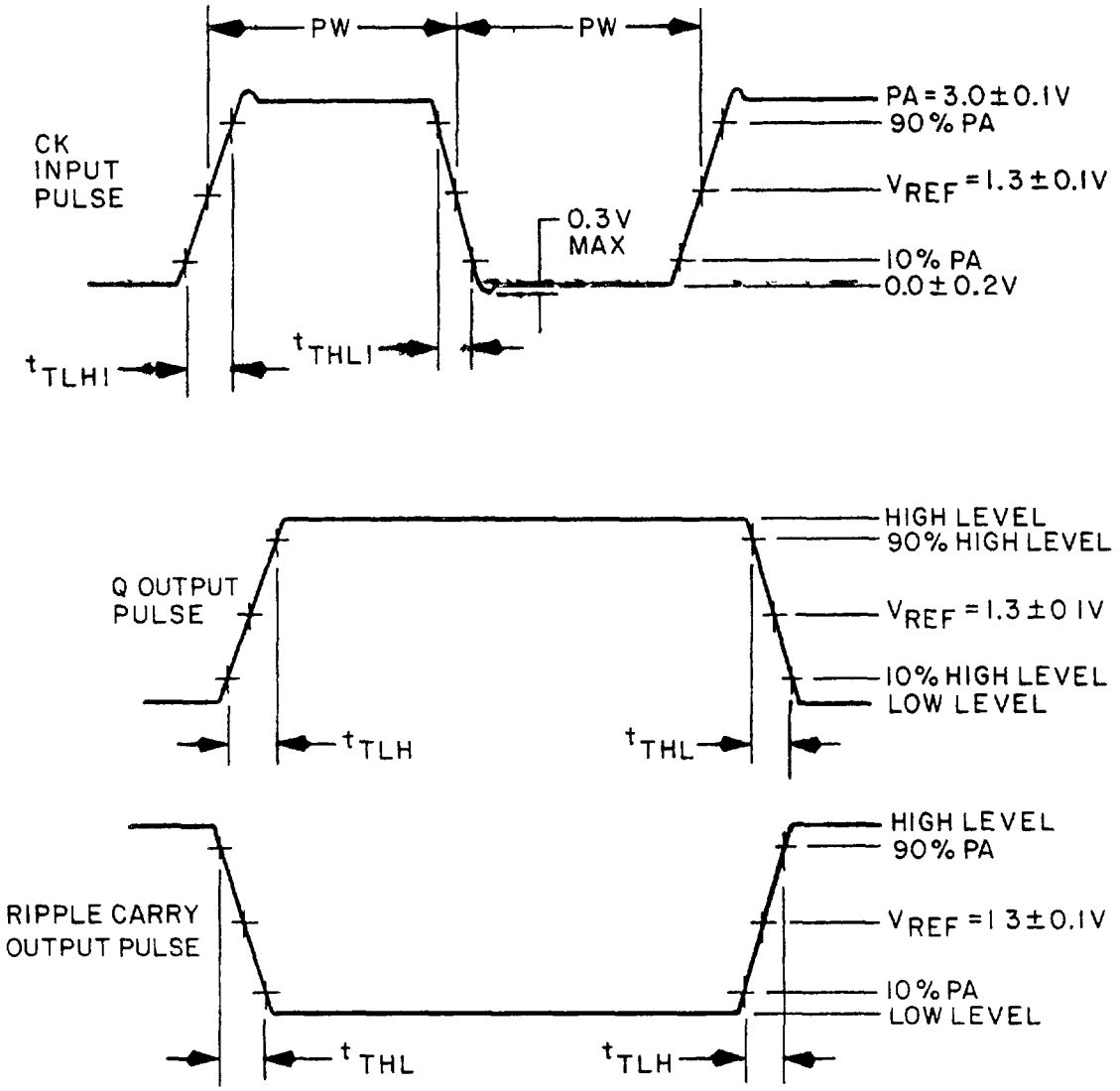
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:43Z
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 1,000 ±10%, 1/2 W.
3. Load all outputs as shown on figure 12.

FIGURE 19. Transition time (t_{TLH} and t_{THL}) test circuit for tables XVI and XVII.

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NOTES:
1. Input pulse parameters:
   PW = 100 ±10 ns.
   $t_{TLH1} = t_{THL1} = 6.0 \pm 2$ ns.
2. Before making measurements, verify that the output low level is $\leq$ 0.4 V maximum and the output high level is $\geq$ 2.5 V minimum.

FIGURE 20. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XVI and XVII.

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TABLE XVIII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R1	CK to 1QA	Propagation delay time (t_{PLH}), C_L = 50 pF	11,21	5	Open	2	24	ns	21,23
R2	CK to 1QB			25					
R3	CK to 1QC			6					
R4	CK to 1QD			26					
R5	CK to 1RCO			24		3	39		
R6	CK to 2QA			34		2	24		
R7	CK to 2QB			14					
R8	CK to 2QC			33					
R9	CK to 2QD			13					
R10	CK to 2RCO			31		3	39		
R11	CK to 3QA			19		2	24		
R12	CK to 3QB			39					
R13	CK to 3QC			18					
R14	CK to 3QD			38					
R15	CK to 3RCO			40		3	39		
R16	1EN$\overline{T}$ to 1RCO		7	24	20	2	19		22,24
R17	2EN$\overline{T}$ to 2RCO		32	31	20		19		
R18	3EN$\overline{T}$ to 3RCO		17	40	20		19		
R19	U/$\overline{D}$ to 1RCO		20	24	7,17,32		29		
R20	U/$\overline{D}$ to 1RCO			24					
R21	U/$\overline{D}$ to 2RCO			31					
R22	U/$\overline{D}$ to 2RCO			31					
R23	U/$\overline{D}$ to 3RCO			40					
R24	U/$\overline{D}$ to 3RCO			40					
R25	CK to 1QA	Propagation delay time (t_{PHL}), C_L = 50 pF	11,21	5	Open	2	27		21,23
R26	CK to 1QB			25					
R27	CK to 1QC			6					
R28	CK to 1QD			26					
R29	CK to 1RCO			24		3	39		

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TABLE XVIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
R30	CK to 2QA	Propagation delay time (t_{PHL}), C_L = 50 pF	11,21	34	Open	2	27	ns	21,23
R31	CK to 2QB			14					
R32	CK to 2QC			33					
R33	CK to 2QD			13					
R34	CK to 2RCO			31		3	39		
R35	CK to 3QA			19		2	27		
R36	CK to 3QB			39					
R37	CK to 3QC			18					
R38	CK to 3QD			38					
R39	CK to 3RCO			40		3	39		
R40	1EN$\overline{T}$ to 1RCO		7	24	20	2	27		22,24
R41	2EN$\overline{T}$ to 2RCO		32	31	20	2	27		
R42	3EN$\overline{T}$ to 3RCO		17	40	20	2	27		
R43	U/$\overline{D}$ to 1RCO		20	24	7,17,32	3	33		
R44	U/$\overline{D}$ to 1RCO			24					
R45	U/$\overline{D}$ to 2RCO			31					
R46	U/$\overline{D}$ to 2RCO			31					
R47	U/$\overline{D}$ to 3RCO			40					
R48	U/$\overline{D}$ to 3RCO			40					
R49	CK to 1QA	Propagation delay time (t_{PLH}), C_L = 150 pF	11,21	5	Open		34		21,23
R50	CK to 1QB			25					
R51	CK to 1QC			6					
R52	CK to 1QD			26					
R53	CK to 1RCO			24		4	49		
R54	CK to 2QA			34		3	34		
R55	CK to 2QB			14					
R56	CK to 2QC			33					
R57	CK to 2QD			13					
R58	CK to 2RCO			31		4	49		

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TABLE XVIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R59	CK to 3QA Propagation delay time (t_{PLH}), C_L = 150 pF	11,21	19	Open	3	34	ns	21,23
R60	CK to 3QB		39					
R61	CK to 3QC		18					
R62	CK to 3QD		38					
R63	CK to 3RCO		40		4	49		
R64	1ENT to 1RCO	7	24	20	2	29		22,24
R65	2ENT to 2RCO	32	31	20	2	29		
R66	3ENT to 3RCO	17	40	20	2	29		
R67	U/D to 1RCO	20	24	7,17,32	3	39		
R68	U/D to 1RCO		24					
R69	U/D to 2RCO		31					
R70	U/D to 2RCO		31					
R71	U/D to 3RCO		40					
R72	U/D to 3RCO		40					
R73	CK to 1QA Propagation delay time (t_{PHL}), C_L = 150 pF	11,21	5	Open		37		21,23
R74	CK to 1QB		25					
R75	CK to 1QC		6					
R76	CK to 1QD		26					
R77	CK to 1RCO		24		4	49		
R78	CK to 2QA		34		3	37		
R79	CK to 2QB		14					
R80	CK to 2QC		33					
R81	CK to 2QD		13					
R82	CK to 2RCO		31		4	49		
R83	CK to 3QA		19		3	37		
R84	CK to 3QB		39					
R85	CK to 3QC		18					
R86	CK to 3QD		38					
R87	CK to 3RCO		40		4	49		

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TABLE XVIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

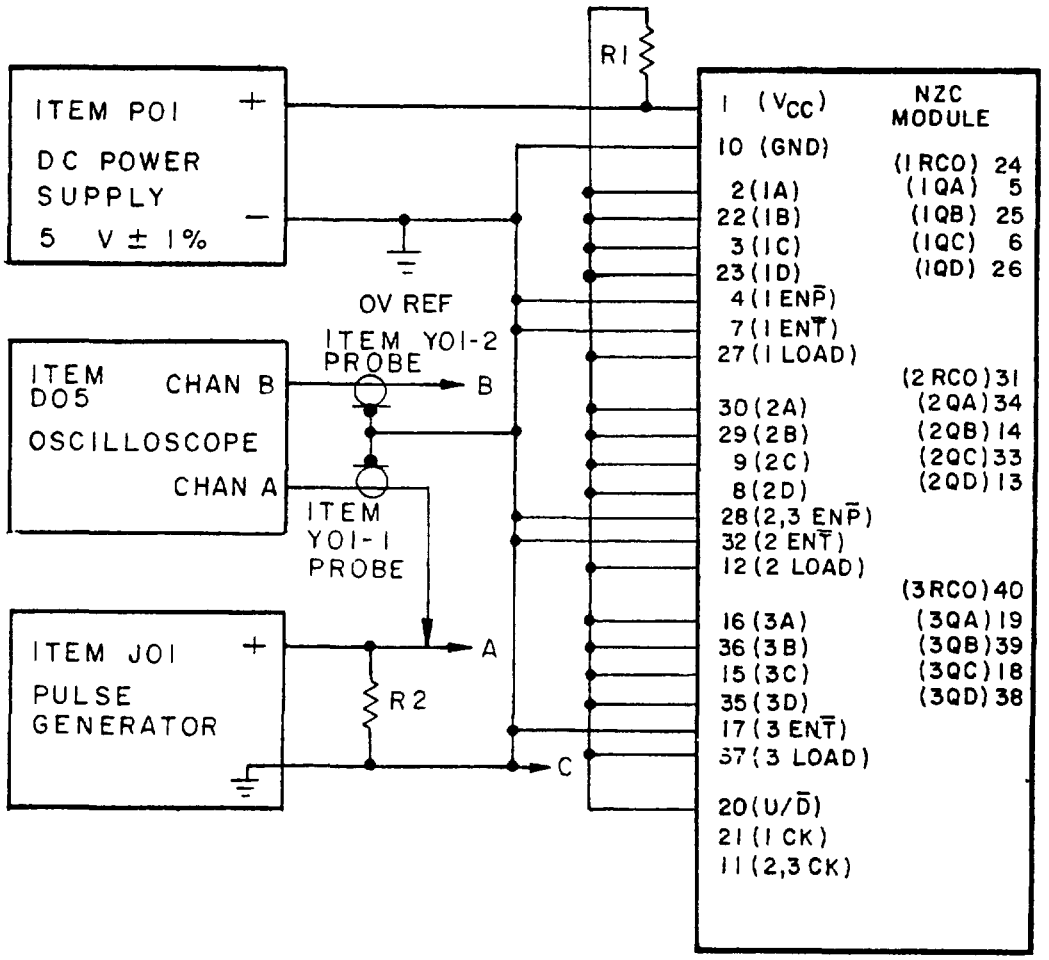
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R88	1EN$\overline{T}$ to 1RCO	7	24	20	3	37	ns	22,24
R89	2EN$\overline{T}$ to 2RCO	32	31	20	3	37		
R90	3EN$\overline{T}$ to 3RCO	17	40	20	3	37		
R91	U/$\overline{D}$ to 1RCO	20	24	7,17,32	4	43		
R92	U/$\overline{D}$ to 1RCO		24					
R93	U/$\overline{D}$ to 2RCO		31					
R94	U/$\overline{D}$ to 2RCO		31					
R95	U/$\overline{D}$ to 3RCO		40					
R96	U/$\overline{D}$ to 3RCO		40					

Propagation delay time (t_{PHL}), C_L = 150 pF

TABLE XIX. Maximum clock frequency check test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
S1	Maximum clock frequency check	26	800	864	ns	25,26
S2	Maximum clock frequency check	13	800	864	ns	25,26
S3	Maximum clock frequency check	38	800	864	ns	25,26

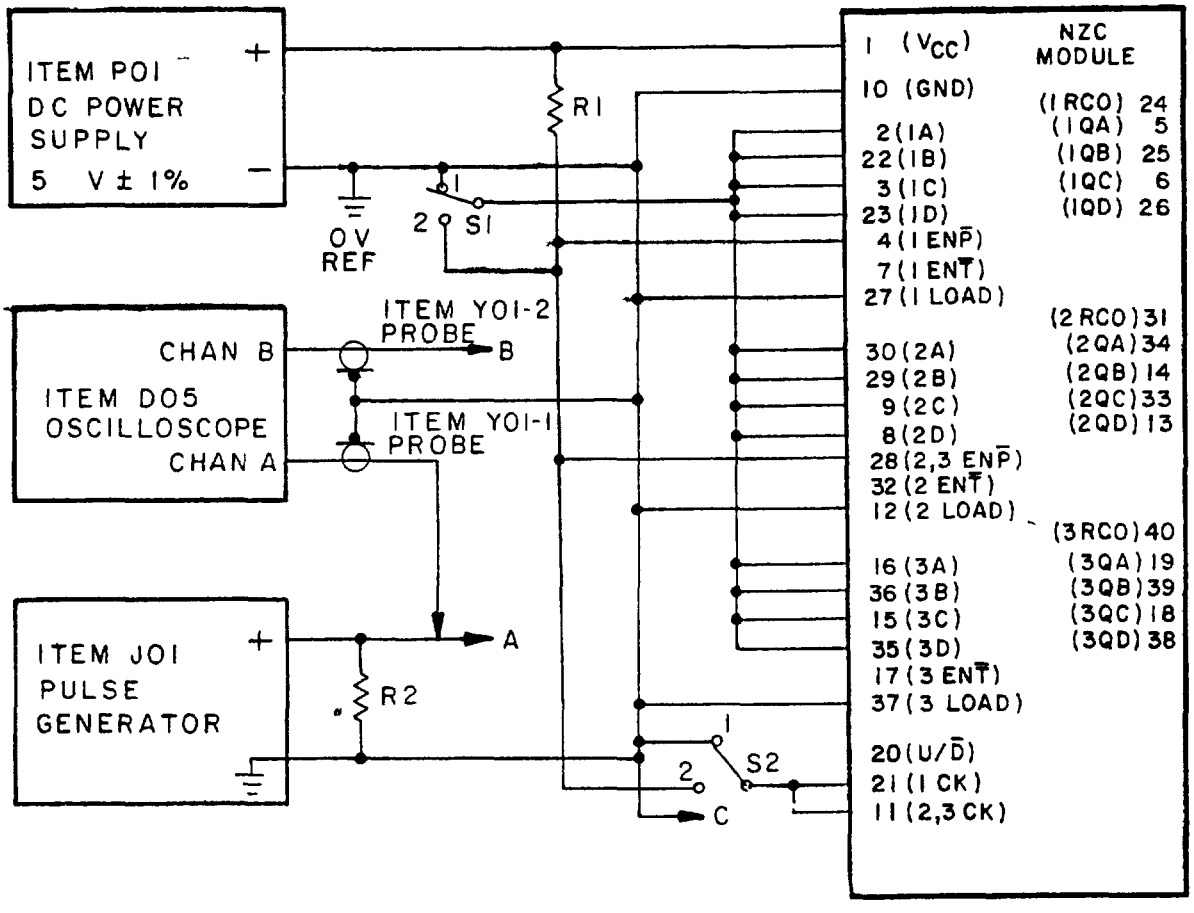
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:43Z
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NOTES:
1. R1 = 6.34 kΩ ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.
3. Load each output as shown on figure 12.

FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVIII.

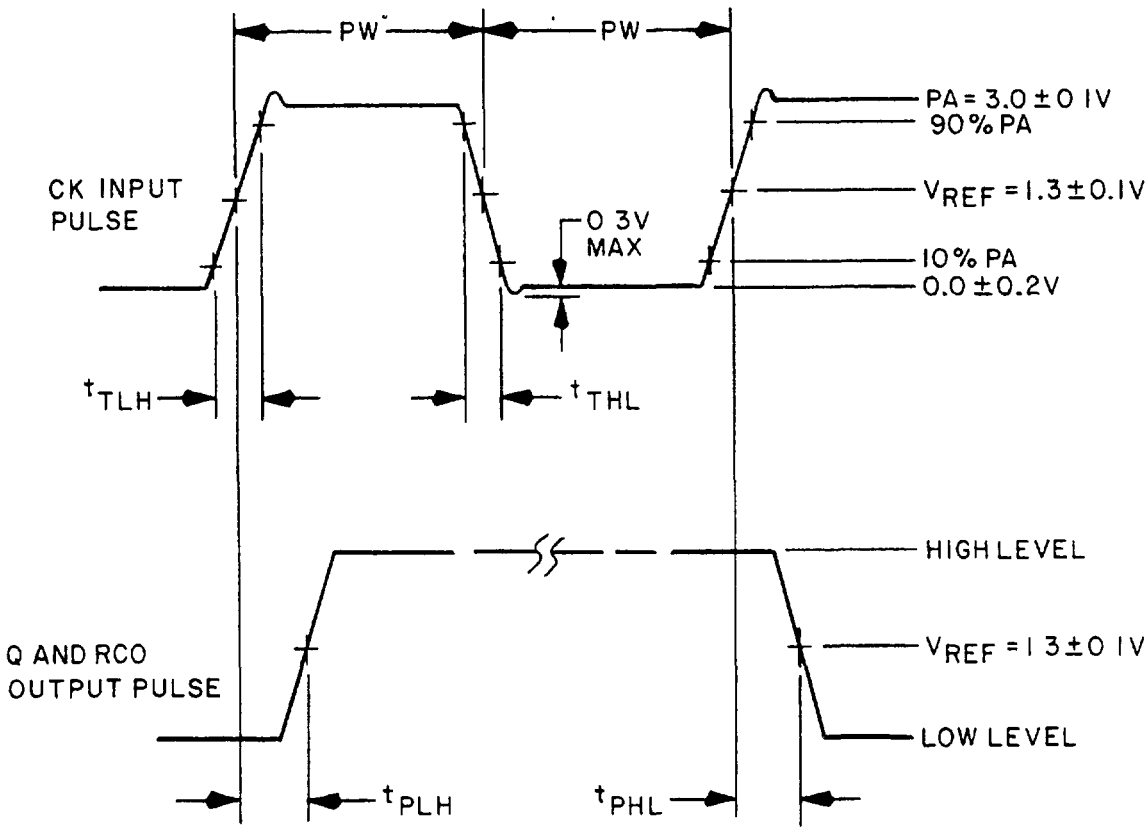
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NOTES:
1. R1 = 6.34 kΩ ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.
3. S1 = S2 = SPDT switch.
4. Before beginning test, momentarily place switch S2 in position 2 then return to position 1.
5. For test conditions R16-R19, R21, R23, R40-R43, R45, R47, R64-R67, R69, R71, R88-R91, R93 and R95 place switch S1 in position 1 and for test conditions R20, R22, R24, R44, R46, R48, R68, R70, R72, R92, R94 and R96 place switch S1 in position 2.
6. Load each output as shown on figure 12.

FIGURE 22. Propgation delay time (t_{PLH} and t_{PHL}) test circuit for table XVIII.

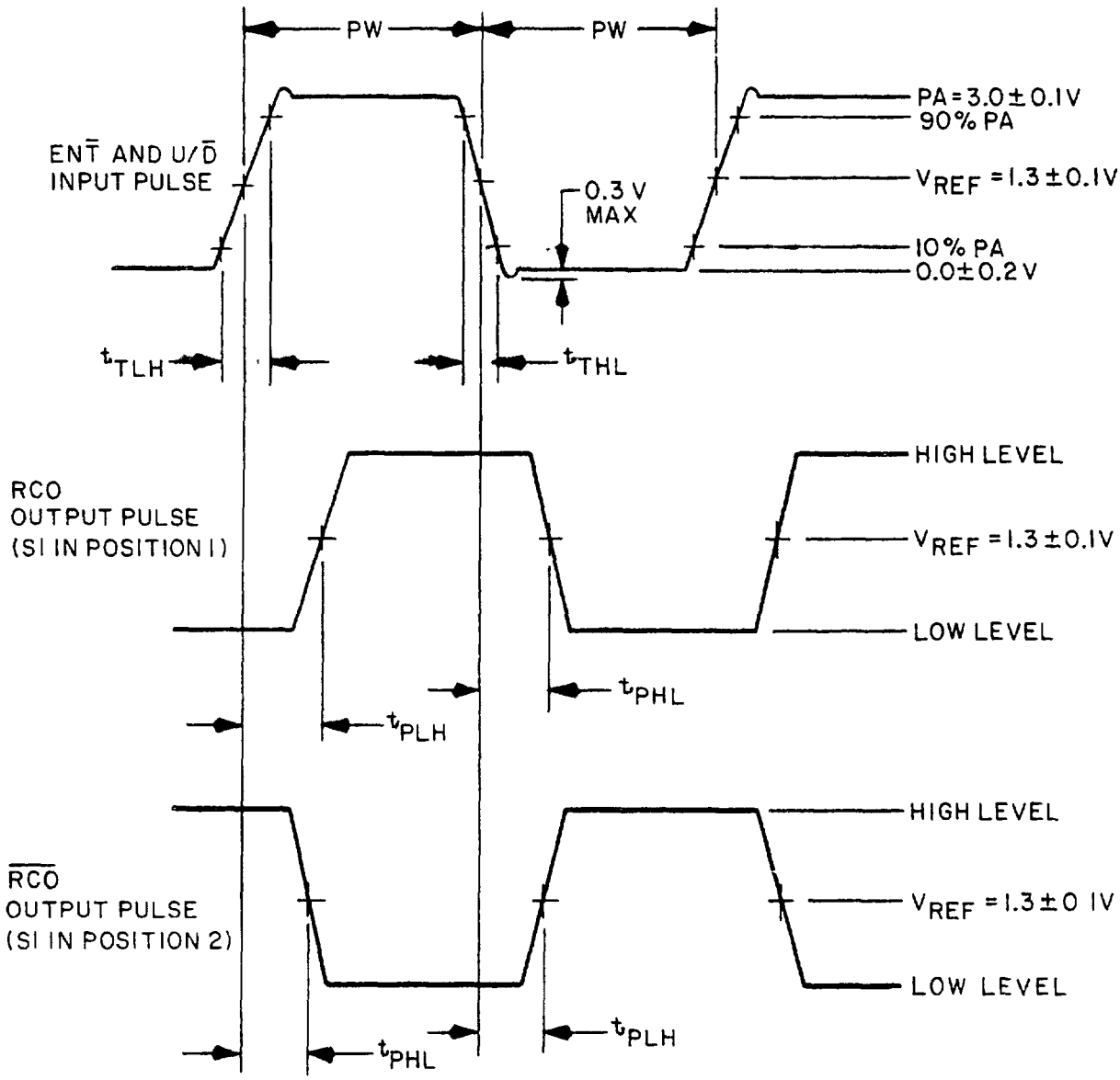
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Check the source to verify that this is the current version before use.



NOTES.
1. Input pulse parameters:
   PW = 100 ±10 ns.
   $t_{TLH} = t_{THL}$ = 6.0 ±2 ns.
2. Before making measurements, verify that the output low level is ≤ 0.4 V maximum and the output high level is ≥ 2.5 V minimum.

FIGURE 23. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVIII.

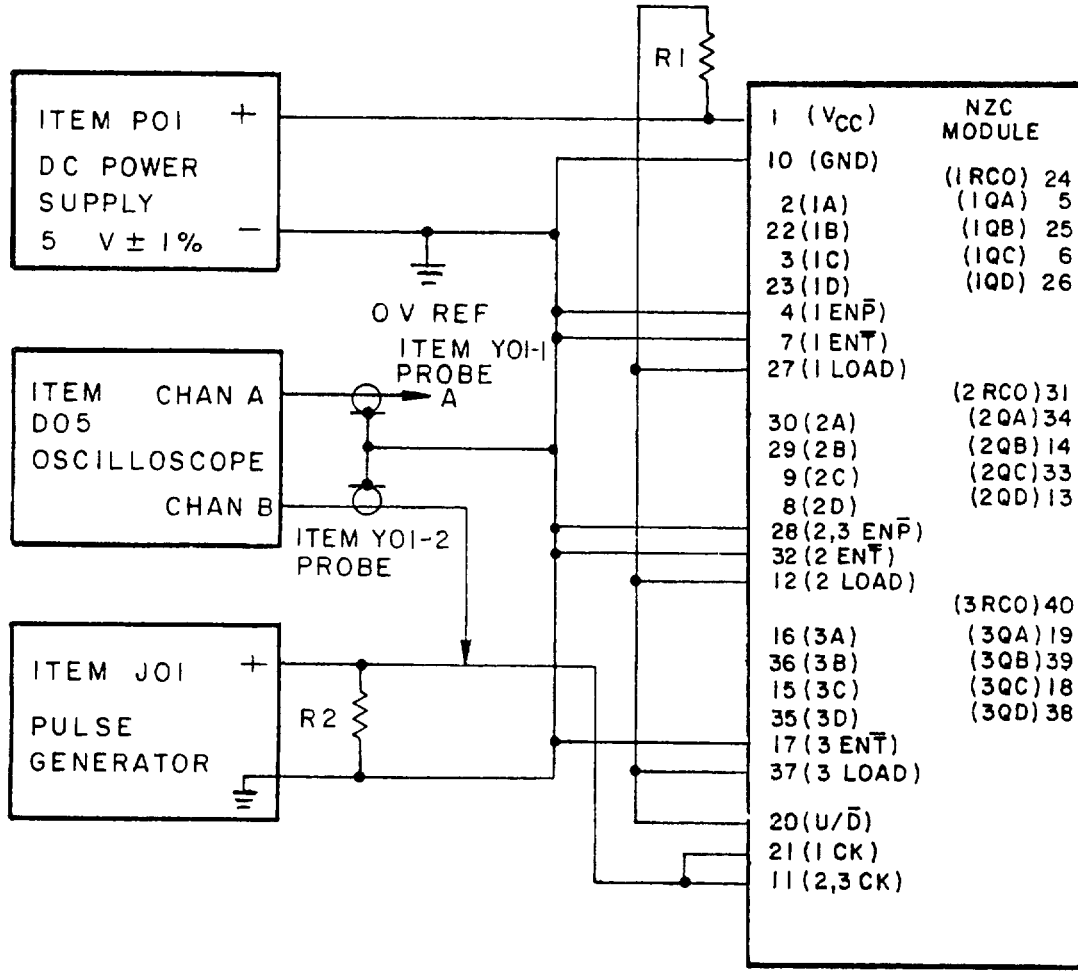
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NOTES:
1. Input pulse parameters:
   PW = 100 ±10 ns.
   $t_{TLH} = t_{THL} = 6.0 \pm 2$ ns.
2. Before making measurements, verify that the output low level is $\leq$ 0.4 V maximum and the output high level is $\geq$ 2.5 V minimum.

FIGURE 24. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVIII.

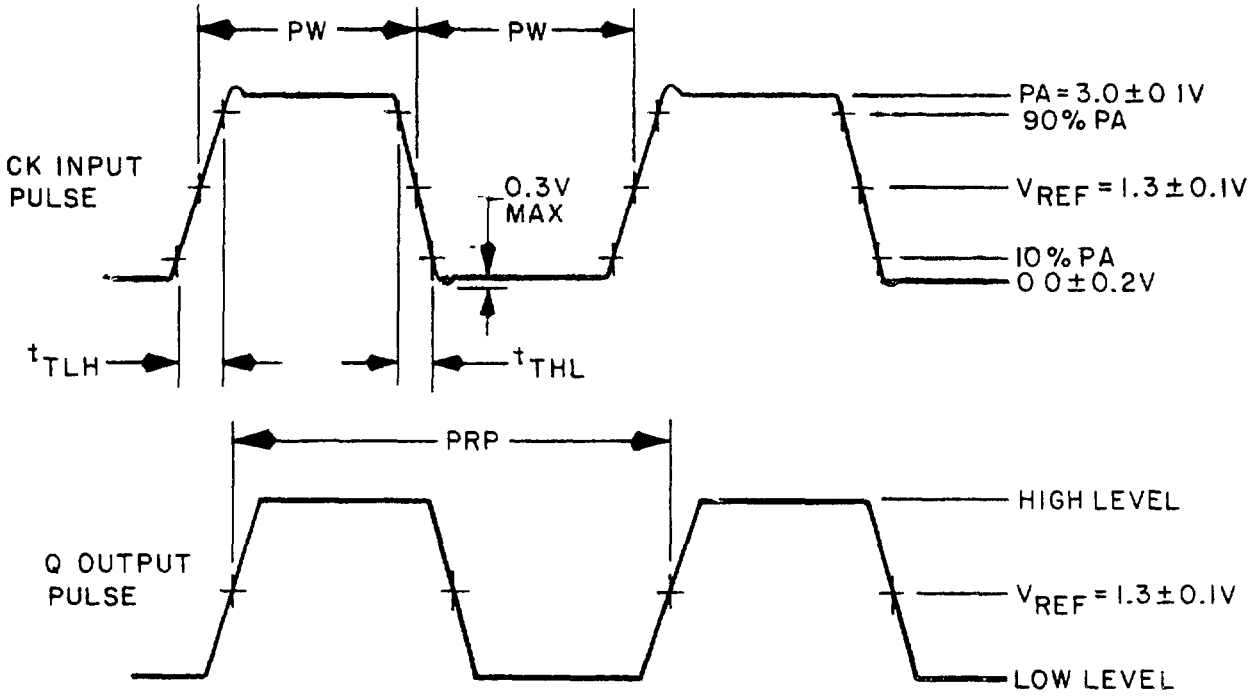
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NOTES:
1. R1 = 6.34 kΩ ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.
3. Load each output as shown on figure 12.

FIGURE 25. Maximum clock frequency check test circuit for table XIX.

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NOTES
1. Input pulse parameters:
   PW = 25 +2, -0 ns.
   $t_{TLH} = t_{THL}$ = 6.0 ±2 ns.
2. Before making measurements, verify that the output low level is $\leq$ 0.4 V maximum and the output high level is $\geq$ 2.5 V minimum.
3. The measurement of the pulse repetition period (PRP) shall meet the requirements of table XIX.

FIGURE 26 Maximum clock frequency check test circuit waveforms for table XIX.

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TABLE XX. Setup time (t_{SUH} and t_{SUL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1A Setup time (t_{SUH})	2	5	12,27,37	4,7,17, 28,32	850	1150	ns	27,28
T2	1B	22	25						
T3	1C	3	6						
T4	1D	23	26						
T5	2A	30	34						
T6	2B	29	14						
T7	2C	9	33						
T8	2D	8	13						
T9	3A	16	19						
T10	3B	36	39						
T11	3C	15	18						
T12	3D	35	38						
T13	1ENP̄	4	5	7,17,32	12,20, 27,37	1700	2300		27,29
T14	2ENP̄	28	34	7,17,32					
T15	3ENP̄	28	19	7,17,32					
T16	1ENT̄	7	5	4,28					
T17	2ENT̄	32	34	4,28					
T18	3ENT̄	17	19	4,28					
T19	1LOAD	27	5	2,4,7,16, 17,28,30, 32	20	850	1150		27,28
T20	2LOAD	12	34		20				
T21	3LOAD	37	19		20				
T22	1U/D̄	20	5		12,27,37				
T23	2U/D̄	20	34		12,27,37				
T24	3U/D̄	20	19		12,27,37				
T25	1A Setup time (t_{SUL})	2	5	12,27,37	4,7,17, 28,32				27,30
T26	1B	22	25						
T27	1C	3	6						

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TABLE XX. Setup time (t_{SUH} and t_{SUL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T28	1D Setup time (t_{SUL})	23	26	12,27,37	4,7,17, 28,32	850	1150	ns	27,30
T29	2A	30	34						
T30	2B	29	14						
T31	2C	9	33						
T32	2D	8	13						
T33	3A	16	19						
T34	3B	36	39	12,27,37					
T35	3C	15	18	12,27,37					
T36	3D	35	38	12,27,37					
T37	1ENP	4	5	7,17,32	12,20,27, 37	1700	2300		27,31
T38	2ENP	28	34	7,17,32					
T39	3ENP	28	19	7,17,32					
T40	1ENT	7	5	4,28					
T41	2ENT	32	34	4,28					
T42	3ENT	17	19	4,28					
T43	1LOAD	27	5	2,4,7,16, 17,28,30, 32	20	850	1150		27,30
T44	2LOAD	12	34		20				
T45	3LOAD	37	19		20				
T46	1U/D	20	5		12,27,37				
T47	2U/D	20	34		12,27,37				
T48	3U/D	20	19		12,27,37				

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TABLE XXI. Hold time (t_{HH} and t_{HL}) test.

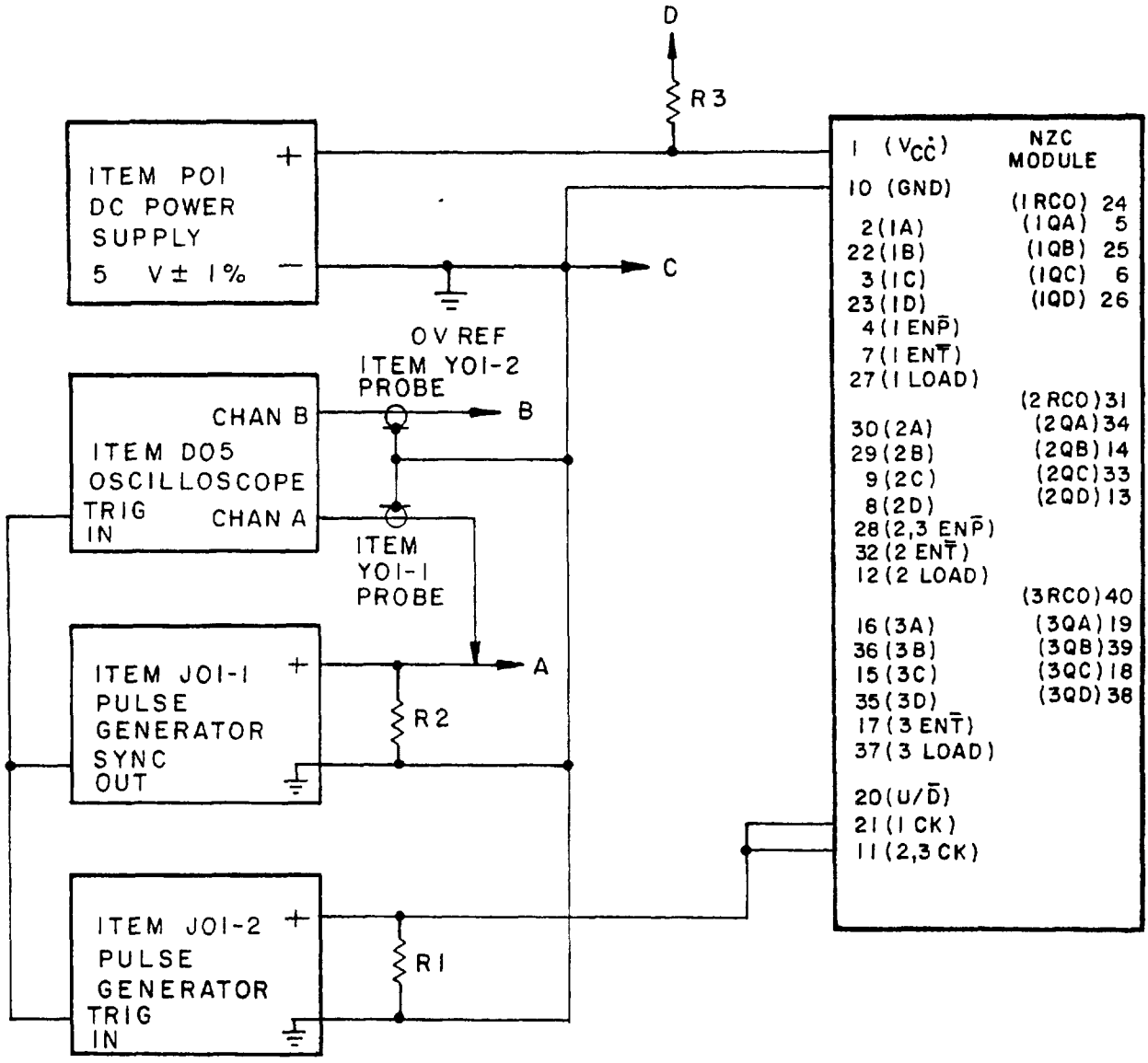
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1A Hold time (t_{HH})	2	5	12,27,37	4,7,17, 28,32	850	1150	ns	27,32
U2	1B	22	25						
U3	1C	3	6						
U4	1D	23	26						
U5	2A	30	34						
U6	2B	29	14						
U7	2C	9	33						
U8	2D	8	13						
U9	3A	16	19						
U10	3B	36	39						
U11	3C	15	18						
U12	3D	35	38						
U13	1ENP̄	4	5	7,17,32	12,20, 27,37	1700	2300		27,33
U14	2ENP̄	28	34	7,17,32					
U15	3ENP̄	28	19	7,17,32					
U16	1ENT̄	7	5	4,28					
U17	2ENT̄	32	34	4,28					
U18	3ENT̄	17	19	4,28					
U19	1LOAD	27	5	2,4,7,16, 17,28,30, 32	20	850	1150		27,32
U20	2LOAD	12	34		20				
U21	3LOAD	37	19		20				
U22	1U/D̄	20	5		12,27,37				
U23	2U/D̄	20	34		12,27,37				
U24	3U/D̄	20	19		12,27,37				

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TABLE XXI. Hold time (t_{HH} and t_{HL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U25	1A Hold time (t_{HL})	2	5	12,27,37	4,7,17, 28,32	850	1150	ns	27,34
U26	1B	22	25						
U27	1C	3	6						
U28	1D	23	26						
U29	2A	30	34						
U30	2B	29	14						
U31	2C	9	33						
U32	2D	8	13						
U33	3A	16	19						
U34	3B	36	39						
U35	3C	15	18						
U36	3D	35	38						
U37	1ENP̄	4	5	7,17,32	12,20,27, 37	1700	2300		27,35
U38	2ENP̄	28	34	7,17,32					
U39	3ENP̄	28	19	7,17,32					
U40	1ENT̄	7	5	4,28					
U41	2ENT̄	32	34	4,28					
U42	3ENT̄	17	19	4,28					
U43	1LOAD	27	5	2,4,7,16, 17,28,30, 32	20	850	1150		27,34
U44	2LOAD	12	34		20				
U45	3LOAD	37	19		20				
U46	1U/D̄	20	5		12,27,37				
U47	2U/D̄	20	34		12,27,37				
U48	3U/D̄	20	19		12,27,37				

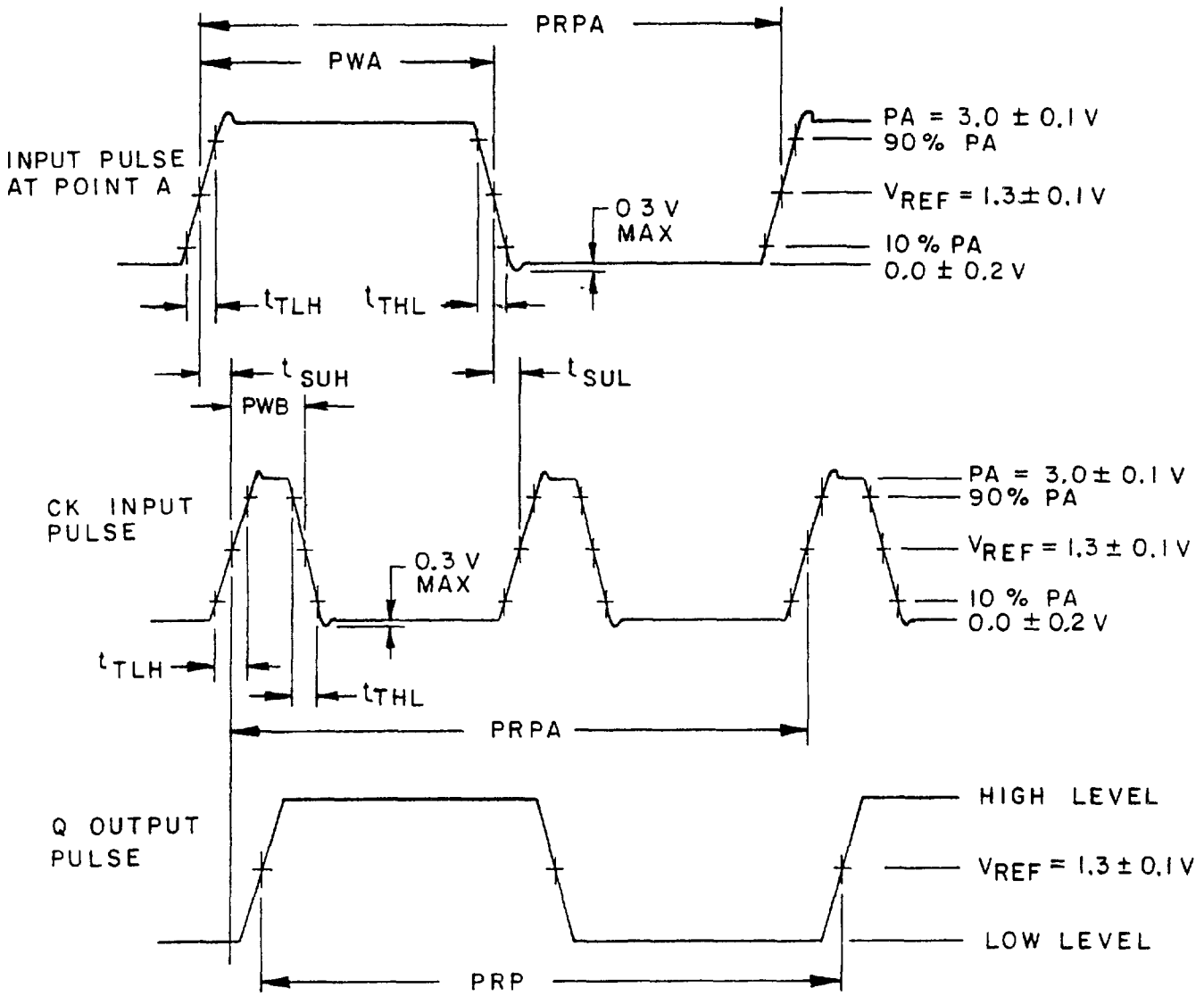
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:43Z
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. R3 = 1 kΩ ±10%, 1/2 W.
3. Set item J01-2 to operate in the dual pulse mode.
4. Load each output as shown on figure 12.

FIGURE 27. Setup and hold time (t_{SUH}, t_{SUL}, t_{HH} and t_{HL}) test circuit for tables XX and XXI.

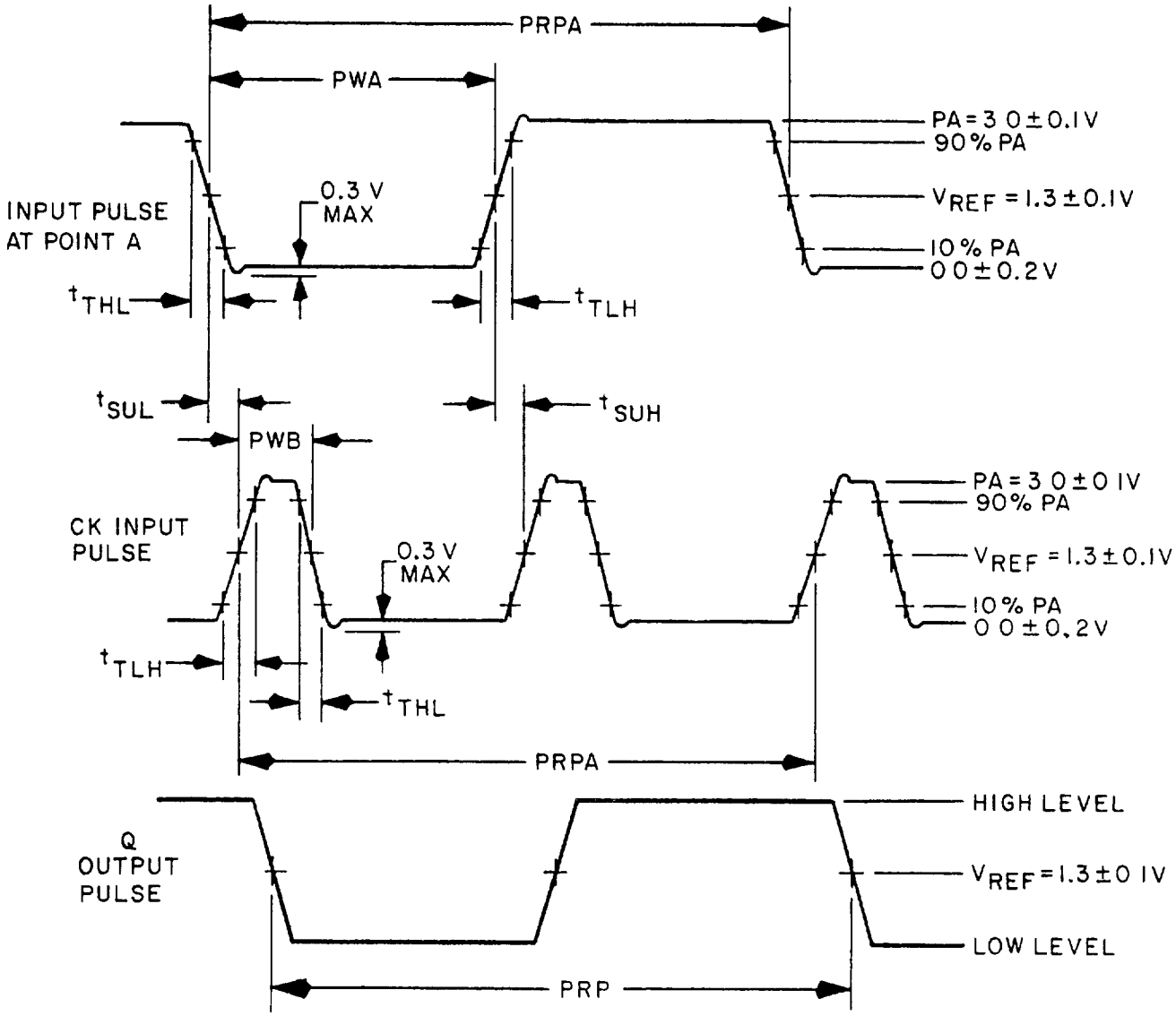
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NOTES
1. Input pulse parameters
   PWA = 500 ±50 ns.
   PWB = 100 ±10 ns.
   PRPA = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 6.0 ±2 ns.
   t_{SUH} = 20 +2, -0 ns for test conditions T1-T12.
   = 25 +2, -0 ns for test conditions T19-T21.
   = 30 +2, -0 ns for test conditions T22-T24.
   t_{SUL} = 50 ±5 ns for test conditions T1-T12 and T19-T24.
2. Before making measurements, verify that the output low level is ≤ 0.4 V maximum and the output high level is ≥ 2.5 V minimum.
3 The measurement of the pulse repetition period (PRP) shall meet the requirements of table XX.

* FIGURE 28. Setup time (t_{SUH}) test circuit waveforms for table XX.

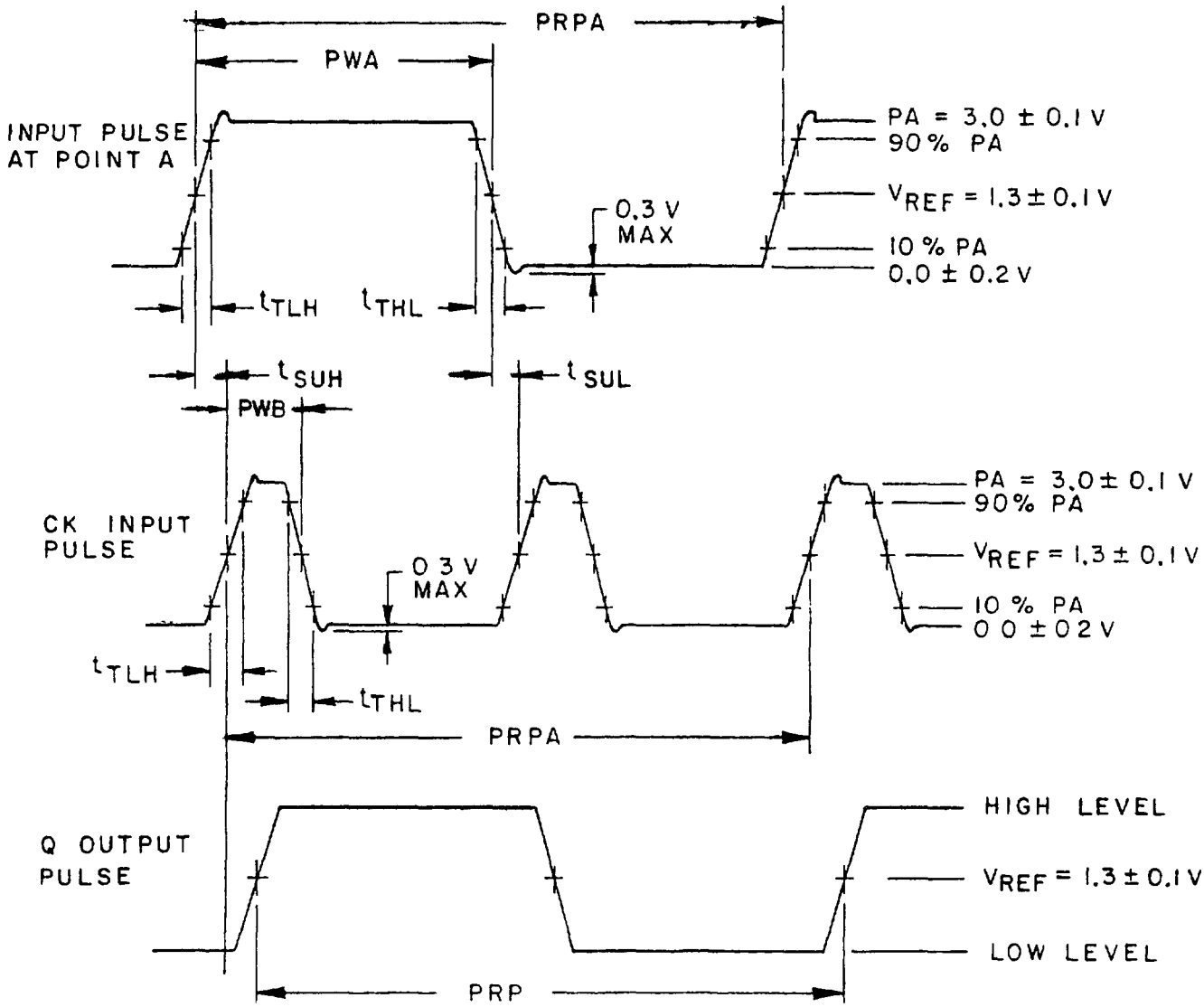
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NOTES.
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 100 ±10 ns.
   PRPA = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 6.0 ±2 ns.
   t_{SUH} = 25 +2, -0 ns for test conditions T13-T18.
   t_{SUL} = 50 ±5 ns for test conditions T13-T18.
2. Before making measurements, verify that the output low level is ≤ 0.4 V maximum and the output high level is ≥ 2.5 V minimum.
3. The measurement of the pulse repetition period (PRP) shall meet the requirements of table XX.

* FIGURE 29. Setup time (t_{SUH}) test circuit waveforms for table XX.

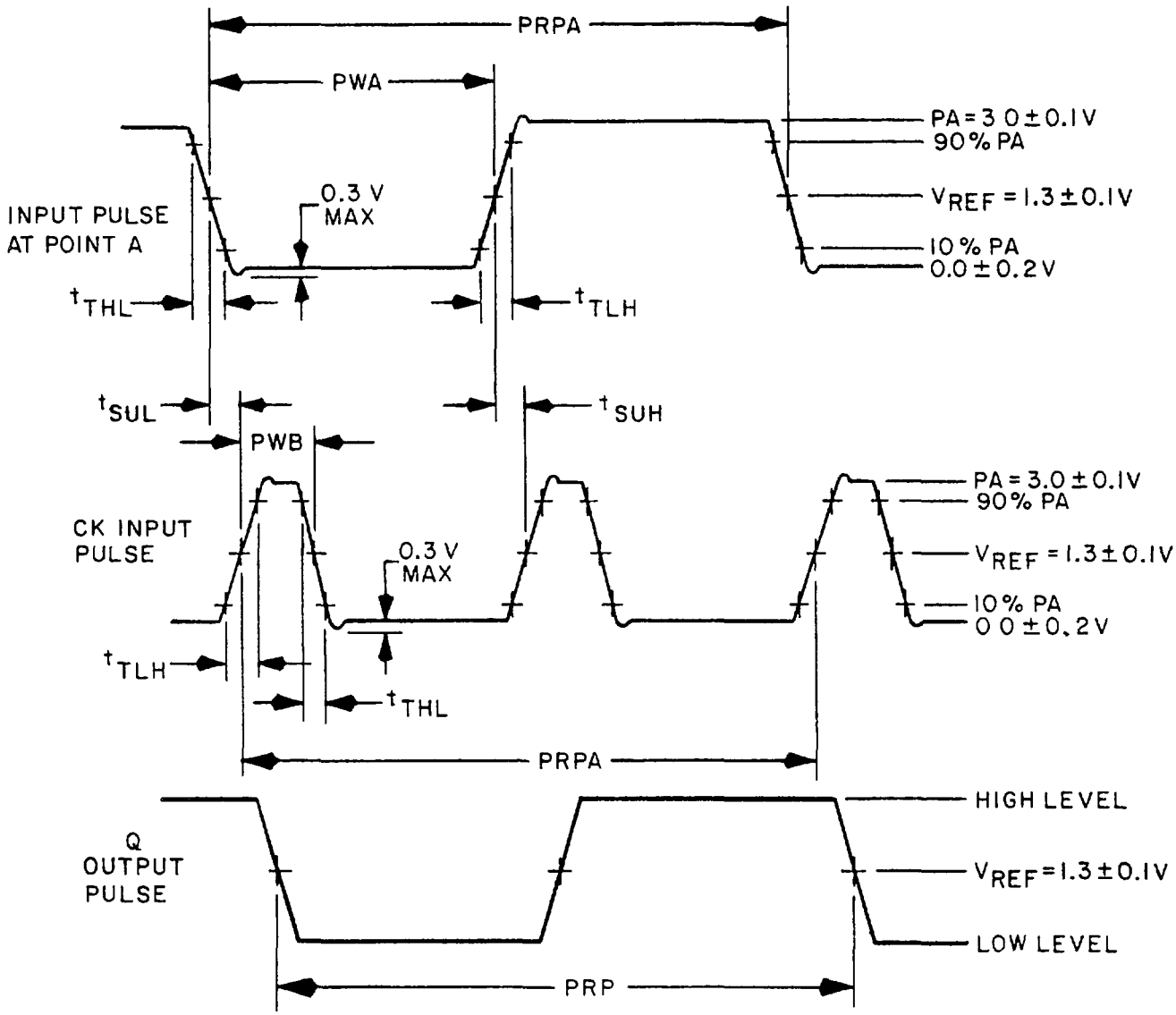
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NOTES:
1. Input pulse parameters.
   PWA = 500 ±50 ns.
   PWB = 100 ±10 ns.
   PRPA = 1,000 ±100 ns.
   $t_{TLH} = t_{THL} = 6.0 \pm 2$ ns.
   t_{SUL} = 20 +2, -0 ns for test conditions T25-T36.
   t_{SUL} = 25 +2, -0 ns for test conditions T43-T45.
   t_{SUL} = 30 +2, -0 ns for test conditions T46-T48.
   t_{SUH} = 50 ±5 ns for test conditions T25-T36 and T43-T48.
2. Before making measurements, verify that the output low level is $\leq$ 0.4 V maximum and the output high level is $\geq$ 2.5 V minimum.
3. The measurement of the pulse repetition period (PRP) shall meet the requirements of table XX.

* FIGURE 30. Setup time (t_{SUL}) test circuit waveforms for table XX.

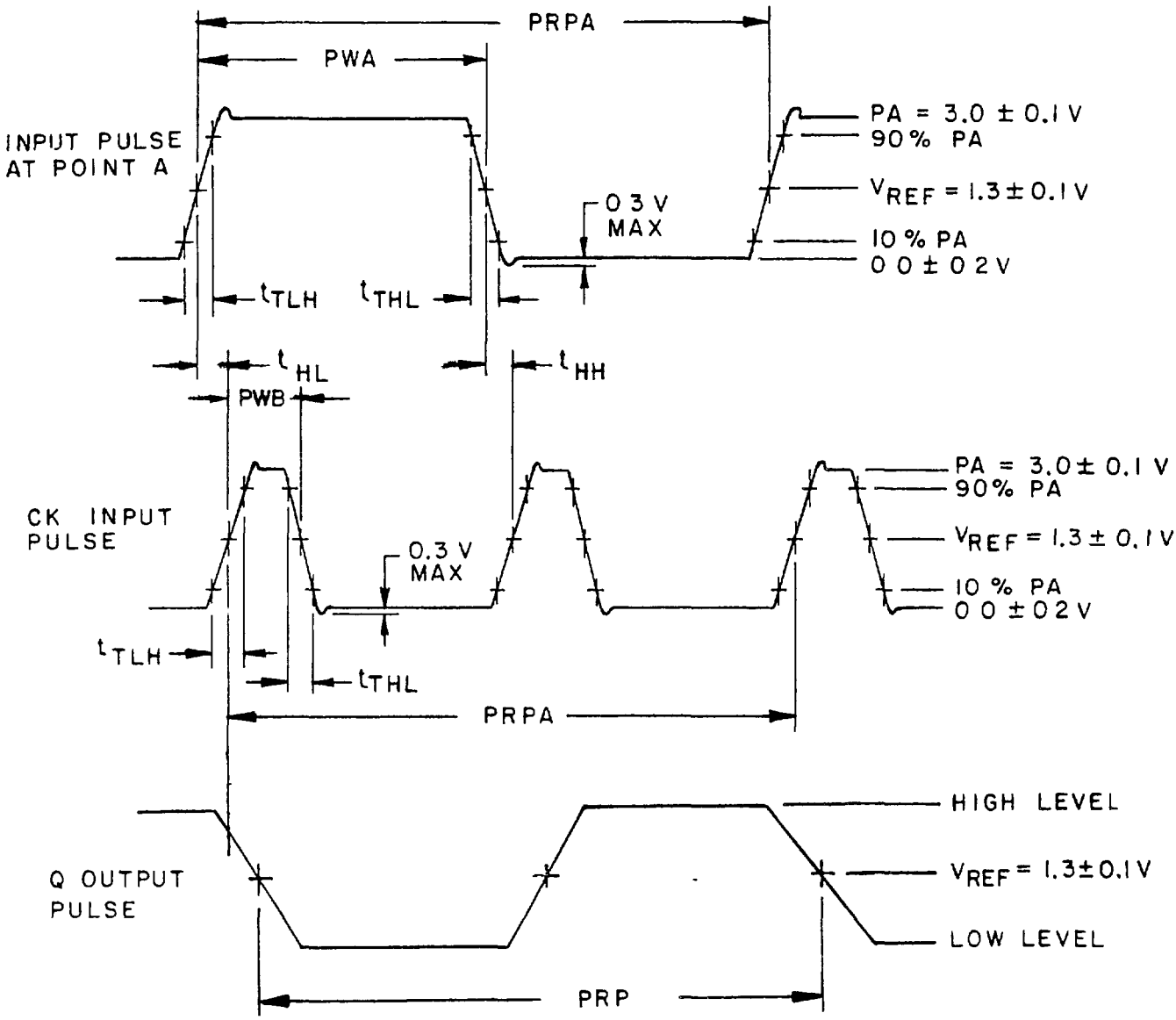
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NOTES.
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 100 ±10 ns.
   PRPA = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 6.0 ±2 ns.
   t_{SUL} = 25 +2, -0 ns for test conditions T37-T42.
   t_{SUH} = 50 ±5 ns for test conditions T37-T42.
2. Before making measurements, verify that the output low level is $\leq$ 0.4 V maximum and the output high level is $\geq$ 2.5 V minimum.
3. The measurement of the pulse repetition period (PRP) shall meet the requirements of table XX.

* FIGURE 31. Setup time (t_{SUL}) test circuit waveforms for table XX.

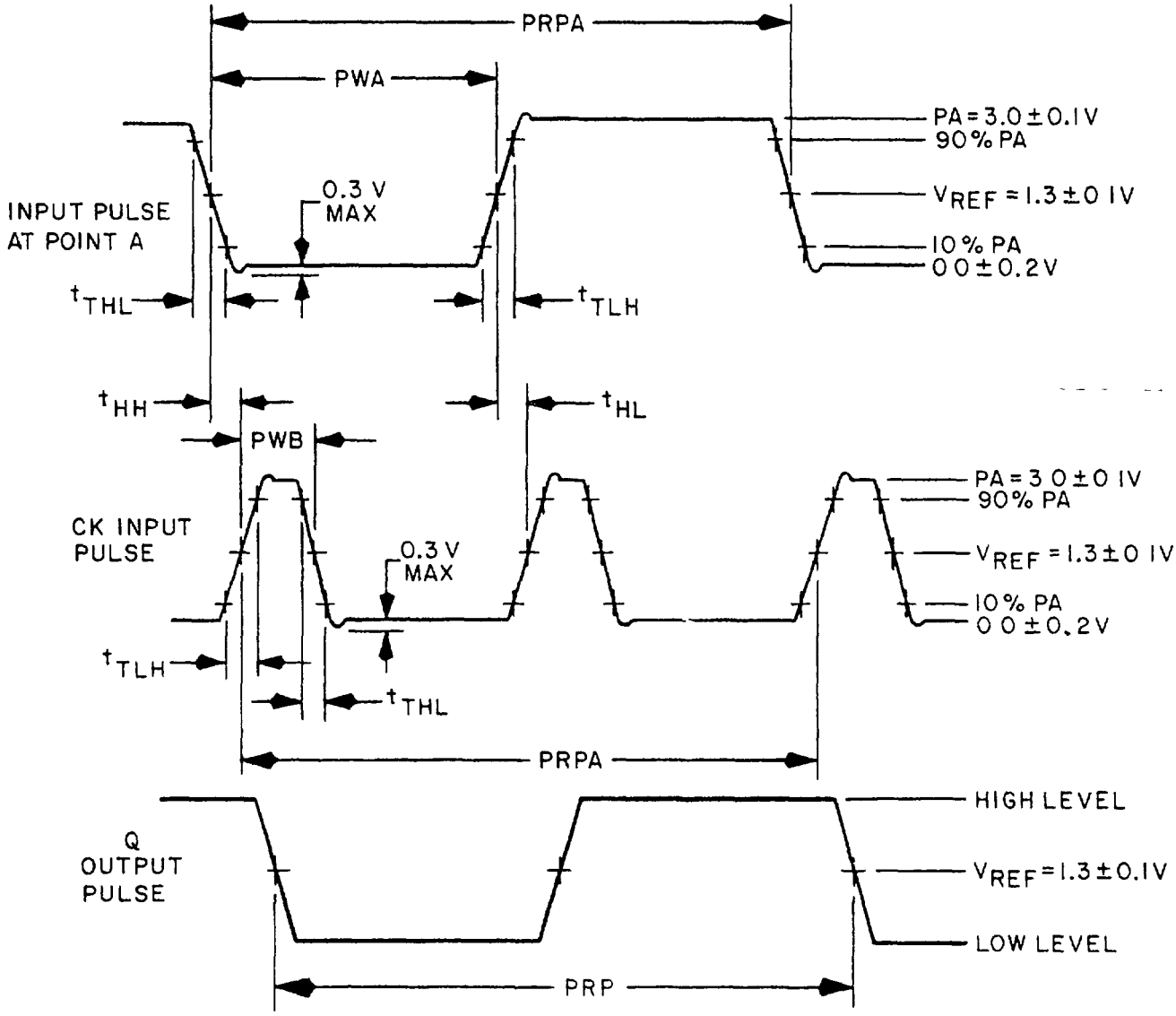
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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 100 ±10 ns.
   PRPA = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 6.0 ±2 ns.
   t_{HH} = 4 +2, -0 ns for test conditions U1-U12 and U19-U24.
   t_{HL} = 10 ±2 ns for test conditions U1-U12 and U19-U24.
2. Before making measurements, verify that the output low level is $\leq$ 0.4 V maximum and the output high level is $\geq$ 2.5 V minimum.
3. The measurement of the pulse repetition period (PRP) shall meet the requirements of table XXI.

* FIGURE 32. Hold time (t_{HH}) test circuit waveforms for table XXI.

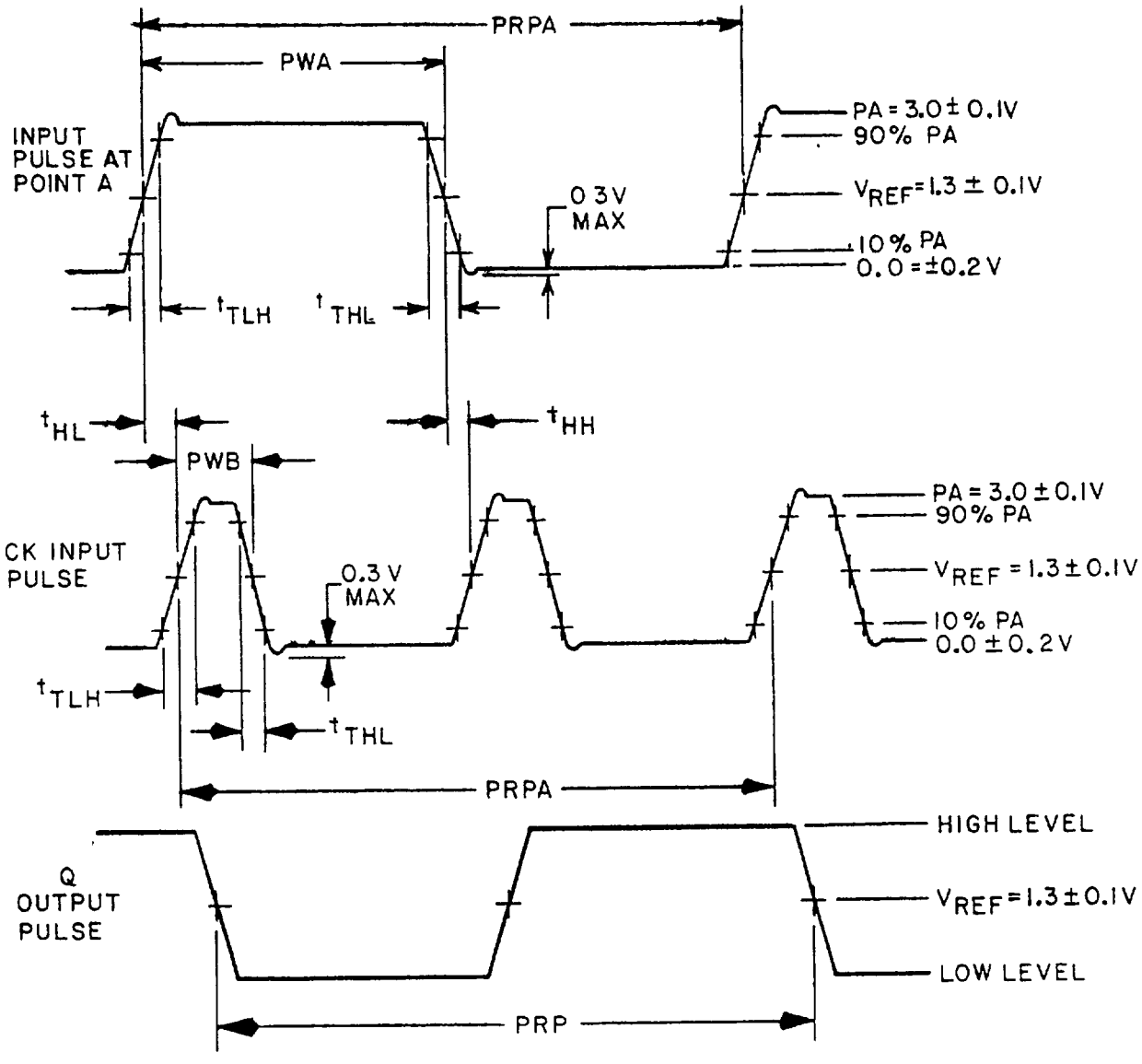
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NOTES:
1. Input pulse parameters
   PWA = 500 ±50 ns.
   PWB = 100 ±10 ns.
   PRPA = 1,000 ±100 ns.
   $t_{TLH} = t_{THL} = 6.0 \pm 2$ ns.
   $t_{HH} = 4 +2, -0$ ns for test conditions U13-U18.
   $t_{HL} = 10 \pm 2$ ns for test conditions U13-U18.
2. Before making measurements, verify that the output low level is $\leq$ 0.4 V maximum and the output high level is $\geq$ 2.5 V minimum.
3. The measurement of the pulse repetition period (PRP) shall meet the requirements of table XXI.

* FIGURE 33. Hold time (t_{HH}) test circuit waveforms for table XXI.

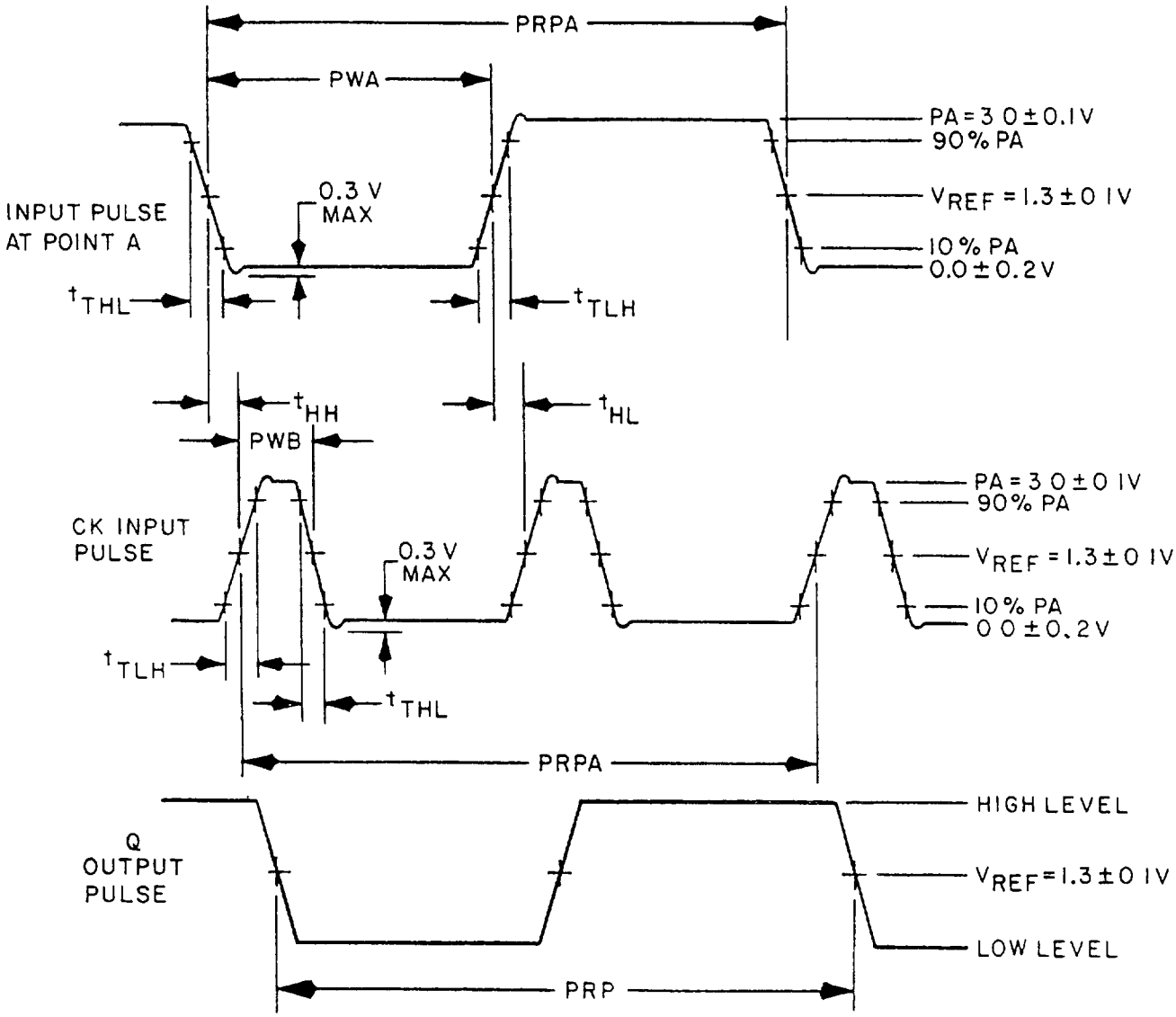
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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 100 ±10 ns.
   PRPA = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 6.0 ±2 ns.
   t_{HL} = 4 +2, -0 ns for test conditions U25-U36 and U43-U48.
   t_{HH} = 10 ±2 ns for test conditions U25-U36 and U43-U48.
2. Before making measurements, verify that the output low level is ≤ 0.4 V maximum and output high level is ≥ 2.5 V minimum.
3. The measurement of the pulse repetition period (PRP) shall meet the requirements of table XXI.

* FIGURE 34. Hold time (t_{HL}) test circuit waveforms for table XXI.

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NOTES
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 100 ±10 ns.
   PRPA = 1,000 ±100 ns.
   $t_{TLH} = t_{THL} = 6.0 \pm 2$ ns.
   t_{HL} = 4 +2, -0 ns for test conditions U37-U42.
   t_{HH} = 10 ±2 ns for test conditions U37-U42.
2. Before making measurements, verify that the output low level is $\leq$ 0.4 V maximum and output high level is $\geq$ 2.5 V minimum
3. The measurement of the pulse repetition period (PRP) shall meet the requirements of table XXI

* FIGURE 35. Hold time (t_{HL}) test circuit waveforms for table XXI.

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TABLE XXII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	1QA Output crosstalk, high level	2	5	Open	3,8,9,15,16,22, 23,29,30,35,36	0	400	mV	36,37
V2	1QB	22	25		2,3,8,9,15,16,23, 29,30,35,36				
V3	1QC	3	6		2,8,9,15,16,22, 23,29,30,35,36				
V4	1QD	23	26		2,3,8,9,15,16,22, 29,30,35,36				
V5	1RCO	Open	24	23	2,3,8,9,15,16,22, 29,30,35,36				
V6	2QA	30	34	Open	2,3,8,9,15,16,22, 23,29,35,36				
V7	2QB	29	14		2,3,8,9,15,16,22, 23,30,35,36				
V8	2QC	9	33		2,3,8,15,16,22, 23,29,30,35,36				
V9	2QD	8	13	Open	2,3,9,15,16,22, 23,29,30,35,36				
V10	2RCO	Open	31	8	2,3,9,15,16,22, 23,29,30,35,36				
V11	3QA	16	19	Open	2,3,8,9,15,22,23, 29,30,35,36				
V12	3QB	36	39		2,3,8,9,15,16,22, 23,29,30,35				
V13	3QC	15	18		2,3,8,9,16,22,23, 29,30,35,36				
V14	3QD	35	38		2,3,8,9,15,16,22, 23,29,30,36				
V15	3RCO	Open	40	35	2,3,8,9,15,16,22, 23,29,30,36				

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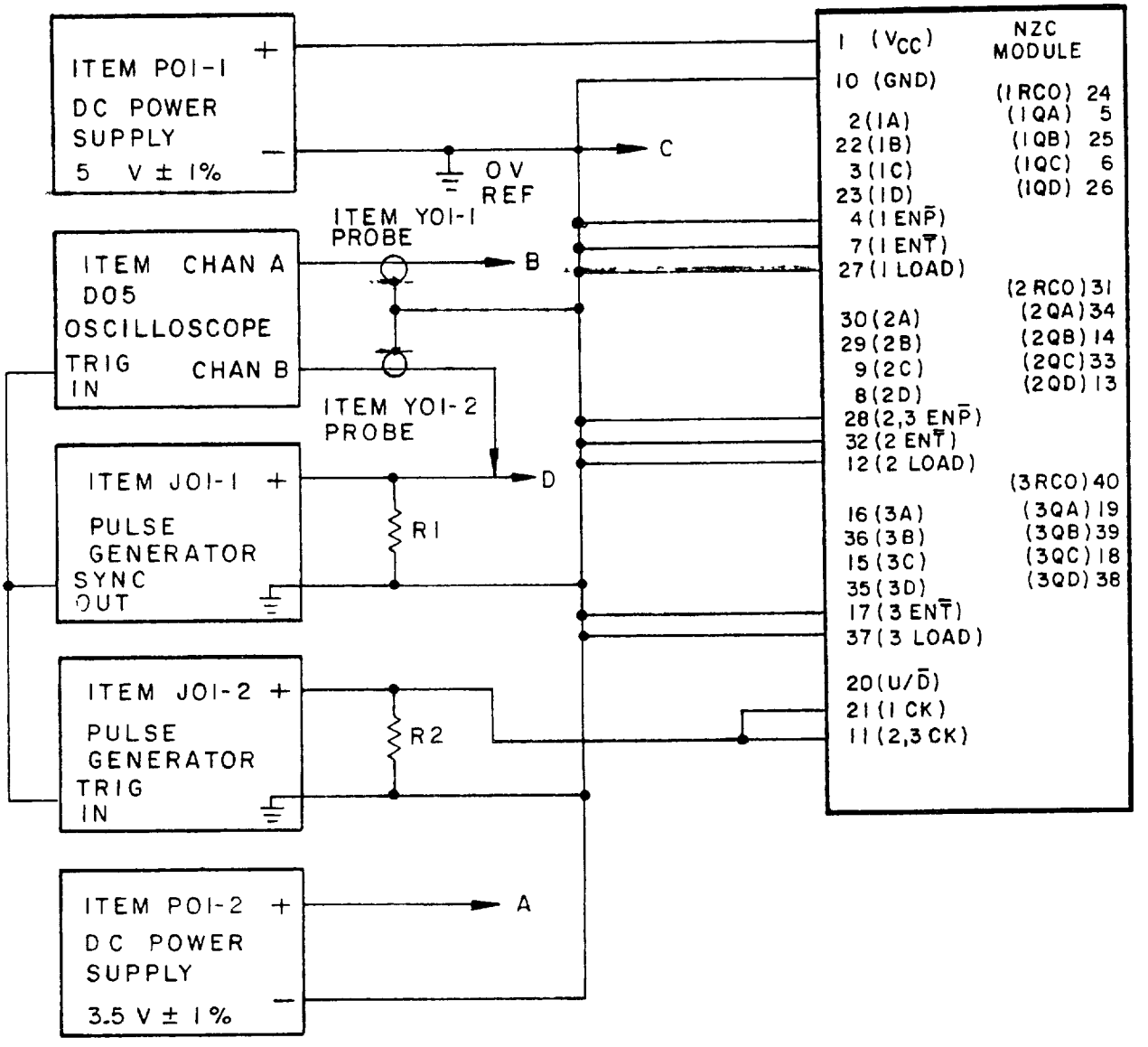
TABLE XXIII. Output crosstalk (low-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
W1	1QA Output crosstalk, low level	Open	5	2	3,8,9,15,16,22, 23,29,30,35,36	0	600	mV	36,37
W2	1QB		25	22	2,3,8,9,15,16,23, 29,30,35,36				
W3	1QC		6	3	2,8,9,15,16,22, 23,29,30,35,36				
W4	1QD		26	23	2,3,8,9,15,16,22, 29,30,35,36				
W5	2QA		34	30	2,3,8,9,15,16,22, 23,29,35,36				
W6	2QB		14	29	2,3,8,9,15,16,22, 23,30,35,36				
W7	2QC		33	9	2,3,8,15,16,22, 23,29,30,35,36				
W8	2QD		13	8	2,3,9,15,16,22, 23,29,30,35,36				
W9	3QA		19	16	2,3,8,9,15,22,23, 29,30,35,36				
W10	3QB		39	36	2,3,8,9,15,16,22, 23,29,30,35				
W11	3QC		18	15	2,3,8,9,16,22,23, 29,30,35,36				
W12	3QD		38	35	2,3,8,9,15,16,22, 23,29,30,36				

TABLE XXIV. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
X	Power supply filter capacitance	1.6	3.0	F	38

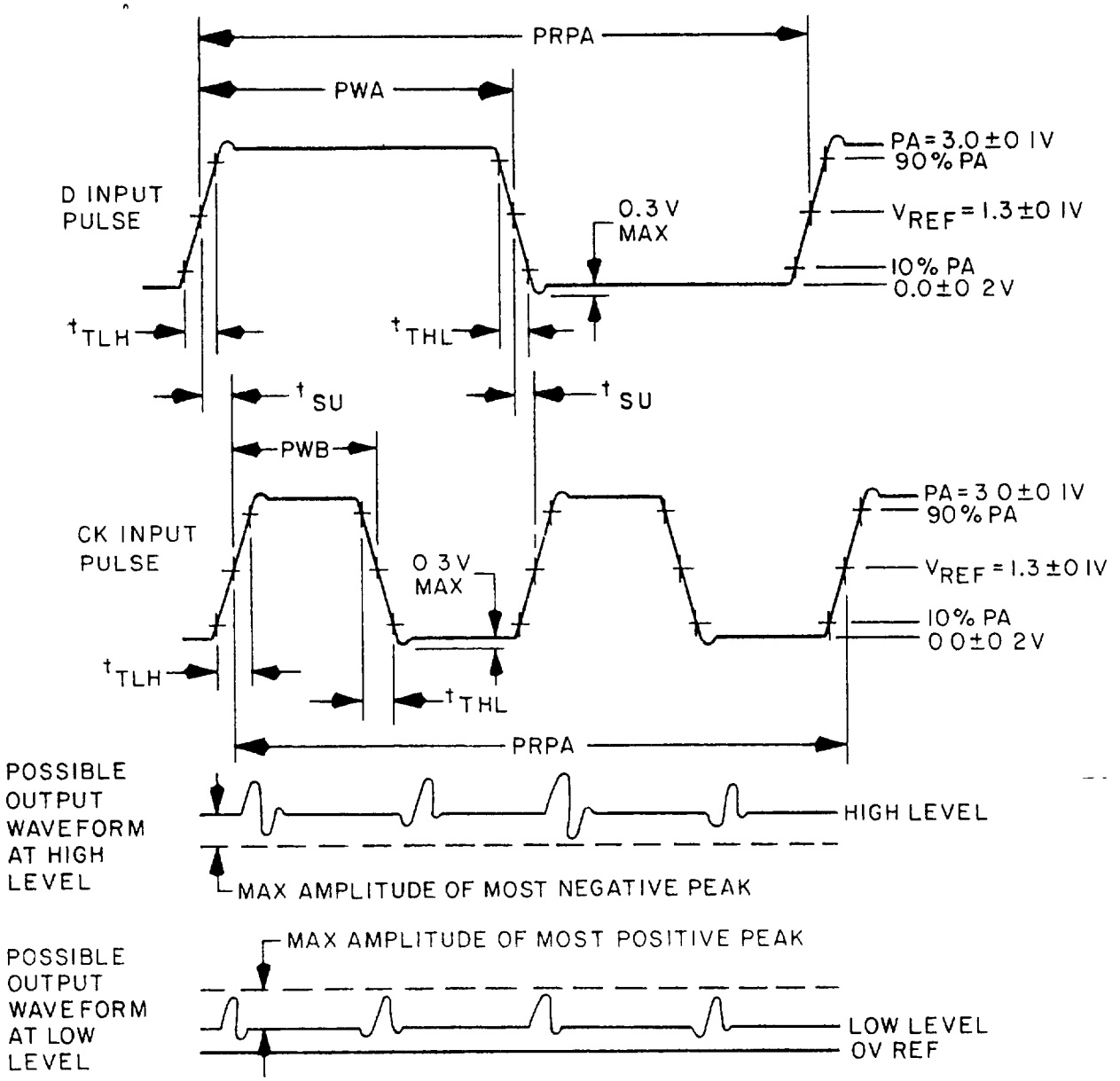
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Load outputs as shown on figure 12.
3. Set item JOI-2 to operate in the dual pulse mode.

FIGURE 36. Output crosstalk (high-level and low-level) test circuit for tables XXII and XXIII.

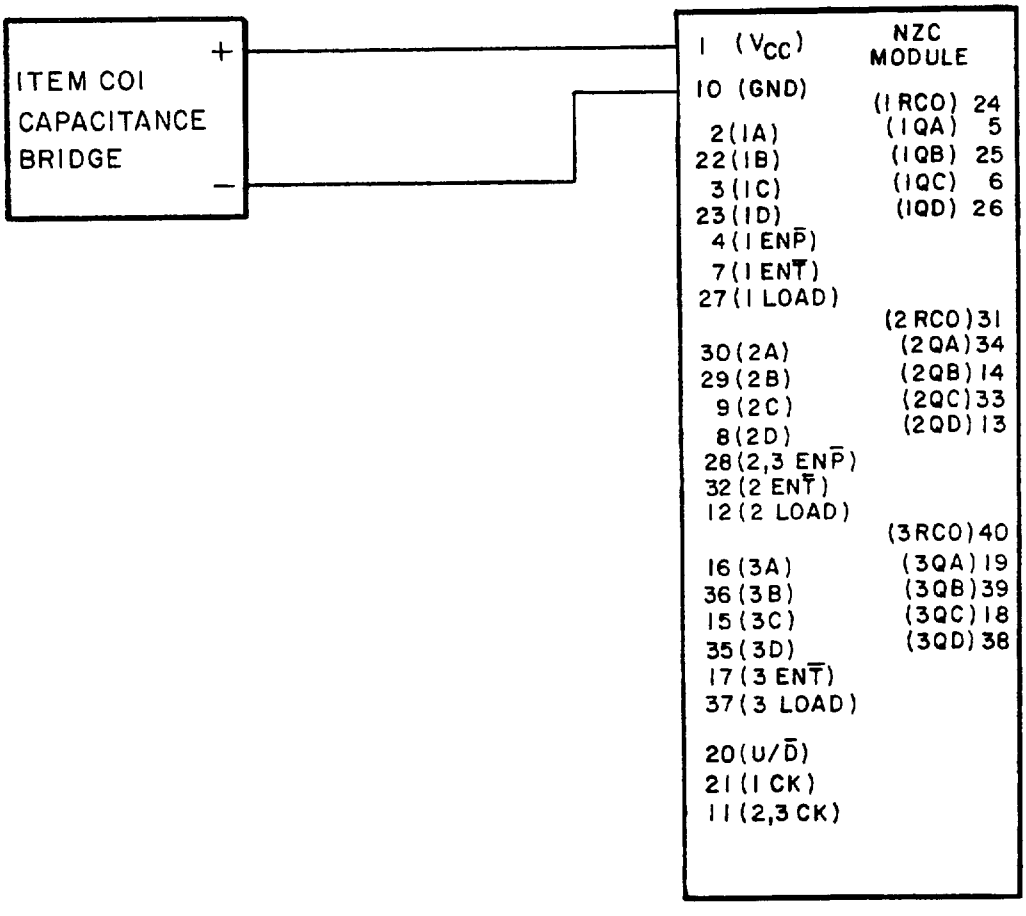
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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 200 ±20 ns.
   $t_{TLH} = t_{THL} = 6.0 \pm 2$ ns.
   PRPA = 1,000 ±100 ns.
   $t_{SU} = 200 \pm 20$ ns
2. The lead length of electrical conductors between terminator resistors R1 and R2 and input pins, probes YO1-1 and YO1-2 to input and output pins, load circuits to output pins, and ground leads of probes to test fixture and module ground pins shall be less than 2 inches (50.8 mm).

FIGURE 37. Output crosstalk (high-level and low-level) test circuit waveforms for tables XXII and XXIII.

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NOTE: Item COI setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

FIGURE 38. Power supply filter capacitance test circuit for table XXIV.

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